

Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies
FIIT-16768-127206

Martin Kubiš
Cross-Lingual Citations Verification Using
Multilingual Transformer Models
Bachelor Thesis

Supervisor: PaedDr. Pavol Batačík

May 2026

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Bachelor Thesis

Study programme: Informatics

Study field: Computer Science

Place of elaboration: Institute of Computer Engineering and Applied Informatics

Supervisor: PaedDr. Pavol Batačík

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BACHELOR THESIS TOPIC

Student: **Martin Kubiš**
Student's ID: 127206
Study programme: Informatics
Study field: Computer Science
Thesis supervisor: PaedDr. Pavol Baťalík
Head of department: doc. Ing. Ján Lang, PhD.

Topic: **A tool for recognizing texts from foreign language translations**

Language of thesis: English

Specification of Assignment:

V prostredí rastúcej dostupnosti zahraničných vedeckých výstupov a rozvoja prekladových technológií je čoraz väčšie riziko zneužitia cudzích odborných textov formou tzv. prekladového plagiátorstva. Tento jav spočíva v preklade vedeckých článkov alebo iných odborných prác z cudzieho jazyka do slovenčiny (alebo iného cieľového jazyka) bez riadneho uvedenia pôvodného autora a zdroja, pričom sa text následne vydáva za vlastný výskum. Takéto konanie je potrebné považovať za plagiát v plnom rozsahu, keďže dochádza k porušeniu autorských práv a etických zásad vedeckej činnosti. V tomto kontexte analyzujte všetky dostupné existujúce riešenia na identifikáciu sémantickej zhody medzi textami. Navrhnite a implementujte vlastnú interpretáciu realizácie rozpoznania sémantickej zhody aspoň pre tri jazyky. Tento rámec pre detekciu preložených plagiátov – napr. porovnávanie významovej podobnosti (semantic similarity), využitie embeddingov (napr. Word2Vec, BERT, Sentence-BERT) alebo spätného prekladu, ktorý umožní porovnanie originálneho textu so suspektným textom v inom jazyku evalvujte nad datasetom netriviálneho obsahu.

Length of thesis: 40

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Assignment of Bachelor thesis approved by: doc. Ing. Ján Lang, PhD. – Study programme supervisor

Declaration of Honour

I declare on my honour that I wrote this thesis independently under the supervision of PaedDr. Pavol Batalík and with the use of the cited literature.

In Bratislava, 11.5.2026

Martin Kubiš

Acknowledgment

I would like to thank my thesis supervisor, PaedDr. Pavol Batalík and Ing. Jakub Dubec, for their guidance, feedback, and support during the preparation of this bachelor's thesis.

Annotation

Slovak University of Technology Bratislava

FACULTY OF INFORMATICS AND INFORMATION TECHNOLOGIES

Study program: Informatics

Author: Martin Kubiš

Bachelor's thesis: Cross-Lingual Citations Verification Using Multilingual
Transformer Models

Supervisor: PaedDr. Pavol Batalík

May, 2026

This bachelor's thesis focuses on cross-lingual citation verification in academic writing. The problem arises when a cited source is written in another language and the citing author reuses the source through translation or paraphrase in a way that may no longer faithfully reflect the original meaning. Conventional plagiarism detection systems are effective mainly for monolingual overlap and are therefore poorly suited to this task. The thesis reviews the context of plagiarism and citation misuse, summarizes related work in cross-lingual plagiarism detection and citation verification, and discusses multilingual transformer models suitable for English-Slovak semantic comparison. The current contribution is a structured proposal for a multi-stage verification pipeline that extracts citations and references from PDF documents, retrieves the cited sources, segments the source text, and compares cited statements with source passages using multilingual embeddings. The thesis also identifies the main risks and limitations of the approach. Special attention is given to practical constraints such as source availability, reference matching, and the quality of text extracted from PDF documents. The proposed solution is intended primarily as a decision-support tool that helps reviewers identify citations requiring closer manual inspection.

Anotácia

Slovenská technická univerzita v Bratislave

FAKULTA INFORMATIKY A INFORMAČNÝCH TECHNOLOGIÍ

Študijný program: Informatika

Autor: Martin Kubiš

Bakalárska práca: Verifikácia citácií vo viacjazyčnom prostredí pomocí viacjazyčných transformerových modelov

Vedúci práce: PaedDr. Pavol Baťalík

Máj, 2026

Táto bakalárska práca sa zameriava na verifikáciu citácií v akademickom písaní vo viacjazyčnom prostredí. Problém vzniká vtedy, keď je citovaný zdroj napsaný v inom jazyku a autor s jeho obsahom ďalej pracuje prostredníctvom prekladu alebo parafrázy tak, že výsledné tvrdenie už nemusí zachytiť pôvodný význam. Bežné systémy na detekciu plagiátorstva sú účinne najmä pri jednojazykovom textovom prekryve, a preto sú na túto úlohu len obmedzene použiteľné. Práca sumarizuje kontext plagiátorstva a nesprávneho citovania, analyzuje súvisiaci výskum v oblasti viacjazyčnej detekcie plagiátorstva a verifikácie citácií a rozoberá viacjazyčné transformerové modely vhodné na sémantické porovnávanie medzi angličtinou a slovenčinou. Hlavným prínosom práce je štrukturovaný návrh viackrokového verifikačného systému, ktorý z PDF dokumentov extrahuje citácie a referencie, získava citové zdroje, segmentuje zdrojový text a porovnáva citované tvrdenia so zdrojovými pasážami pomocou viacjazyčných embeddingov. Práca zároveň identifikuje hlavné riziká a limity navrhnutého prístupu. Osobitná pozornosť je venovaná praktickým obmedzeniam dostupnosti zdrojov, párovania referencií a kvalite textu extrahovaného z PDF dokumentov. Navrhované riešenie je určené predovšetkým ako podporný nástroj pre hodnotiteľa, ktorý pomáha odhaliť citácie vyžadujúce podrobnejšiu manuálnu kontrolu.

Contents

1	Technical Abstract	1
2	Lay Summary	1
3	Introduction	2
4	Problem Definition	3
5	Overview of Related Solutions	4
6	Solution Overview	6
6.1	Architecture and workflow	6
6.1.1	Document understanding and reference alignment	8
6.1.2	Source retrieval and evidence preparation	8
6.1.3	Semantic verification	9
6.2	Runtime architecture and external services	9
6.3	Reporting and limitations	10
7	Risk Assessment	11
8	Experimental Reproducibility and Impact	12
8.1	Dataset and experimental scope	12
8.2	Hardware and runtime environment	12
8.3	Reproducibility and open science	13
8.4	Reference corpus characteristics	14
8.5	Reference availability and retrieval results	15
8.6	Similarity analysis results	18
8.7	Interpretation and limitations	19
9	Sustainability and Environmental Impact	20
10	Employability	20
11	Teamwork	21
12	Conclusions	21
13	Resumé	22
	References	25

1 Technical Abstract

This bachelor's thesis addresses the problem of cross-lingual citation verification in academic writing. The problem arises when an author cites a source written in another language and then reuses its content through translation or paraphrase. In such cases, standard plagiarism detection systems often fall short because they focus mainly on surface overlap and are not designed to assess whether the cited claim is semantically supported by the original source. This leaves a gap between plagiarism detection and citation verification.

The thesis outlines the legal and ethical context of plagiarism, reviews prior work on cross-lingual plagiarism detection and citation verification, and compares multilingual transformer-based approaches suitable for semantic comparison across languages.

The main result is a proposal for a multi-stage verification pipeline. The proposed system takes a PDF document as input, extracts in-text citations and bibliography entries, retrieves the cited source documents, segments the retrieved text, and compares cited claims with source passages using multilingual embeddings. The intended output is a protocol-style report that supports human assessment by highlighting the passages most relevant to each citation.

2 Lay Summary

This thesis is about checking whether academic citations are actually used honestly when the original source is written in a different language. A student may cite an English paper in a Slovak text, translate the idea, and slightly rewrite it. Even if a citation is present, the rewritten sentence may not match what the source really says. Ordinary plagiarism tools usually compare words and phrases, so they often miss this kind of problem.

The goal of the thesis is to explore how a computer system could help with this task. The proposed tool would read a thesis in PDF form, detect the citations and the bibliography, try to find the original sources, and then compare the cited statements with the source text. To do that, it would use multilingual language models that can compare meaning across languages rather than just identical wording.

3 Introduction

Academic writing depends on citations not only for attribution but also for verification. A reader should be able to follow a reference and confirm that the cited source really supports the claim made in the text. Recent studies show that this link between claim and source is not always reliable: a citation may be present, yet the cited work may only partially support the statement, be interpreted incorrectly, or be used in a misleading way[1][2][3].

In the Slovak academic environment, this problem is grounded both in law and in university practice. The Higher Education Act[4] requires that a submitted thesis be, at least in part, the result of the student's own independent work. The Copyright Act[5] protects original works as results of creative intellectual activity and grants the author both moral and economic rights, including the right to be identified as the creator and to control the use of the work.

This problem overlaps with plagiarism, but it is not identical to it. Plagiarism is commonly defined as presenting another person's writing as one's own, a form of misconduct closely connected with forgery and piracy[6]. In academic practice, however, the issue goes beyond simple copy-paste. It also includes uncredited reuse of ideas, misleading paraphrase, false attribution, and other forms of source misuse discussed in the plagiarism-detection literature[7]. Anderson and Steneck[8] argue that the ethical problem does not arise from learning from prior work, but from presenting others' ideas or words as one's own without proper acknowledgment. Citation misuse is one important part of that broader problem because a source can be formally acknowledged and still be used in a misleading way.

This thesis focuses on a specific form of source misuse that becomes especially difficult when the source and the citing text are written in different languages. An author may cite a foreign-language source, translate or paraphrase its content, and still provide a formally correct reference. In such a case, the central problem is not missing attribution, but whether the cited source actually supports the claim made in the new text. This places the work between two related research areas. Cross-lingual plagiarism detection studies reuse across languages, including direct translation and paraphrased translation[9], while citation verification asks whether a cited source actually supports the statement to which it is attached[10]. The task addressed in this thesis combines both perspectives: the wording may change across languages, yet the evidential relationship between claim and source must still remain accurate.

4 Problem Definition

The core problem addressed in this thesis is not whether a citation is present, but whether the cited source semantically supports the claim attached to it. In a typical scenario, an author cites an English article in a Slovak thesis, translates the relevant claim, paraphrases it, and presents the result with a formally correct reference. The citation may look proper on the surface, yet the translated statement may oversimplify, distort, or even contradict the original meaning. This kind of misuse is harder to detect than literal copying because the wording changes while the argumentative role of the source remains central[9].

The articles on citation accuracy help define this problem more precisely. Mogull distinguishes between minor inaccuracies and major content errors[1]. In the major cases, the referenced source fails to substantiate, is unrelated to, or directly contradicts the cited assertion. Pavlovic et al.[2] use a similar support-based criterion, identifying error types such as nonexistent findings, incorrect interpretation of results, and chains of repeated citation errors. This thesis is primarily concerned with these claim-support failures, not with typographical mistakes in the reference list or other purely bibliographic defects.

This problem belongs to the broader domain of plagiarism and source misuse, which is not limited to direct copy-paste behaviour. A widely cited taxonomy distinguishes deliberate copy-paste plagiarism, simple paraphrasing, patchwork paraphrasing, self-plagiarism, illegitimate-source plagiarism, retweet plagiarism, metaphor plagiarism, and idea plagiarism[11].

Cross-lingual plagiarism itself is usually defined as reuse in which a suspicious document in one language originates from a document in another language, either through exact translation or paraphrased translation[12]. Ehsan et al.[12] group the main comparison strategies into syntax based, dictionary based, parallel corpus based, comparable corpus based, and multilingual semantic network based approaches.

The practical importance of the problem is considerable. Foltýnek et al.[13] note that “even in the best case, i.e., if the plagiarism is discovered, reviewing and punishing plagiarized research papers and grant applications still causes a high effort for the reviewers, affected institutions, and funding agencies”. If it remains undiscovered, plagiarists may gain academic credit, funding, or career advancement based on misappropriated work[13]. In the case of citation verification, the risk is not only theft of text but also the spread of unsupported or misleading claims in academic writing. Mehregan[3] further warns that some unsupported references

may be intentional citation manipulation, used to increase citation counts rather than to support the argument. A system that can automatically assist with cross-lingual citation checking would therefore address both integrity and quality-control concerns.

The goal of this thesis is not to solve general plagiarism detection. Instead, it aims to define a focused verification pipeline for academic texts, in which the system identifies citations, retrieves sources, and estimates whether the cited source semantically supports the corresponding claim.

5 Overview of Related Solutions

The most mature neighbouring area is cross-lingual plagiarism detection. Early work relied on dictionary-based translations[14] and syntax-based approaches[15], but these tend to be computationally expensive and limited by available resources. Classical retrieval-oriented or lexical methods, including string matching[16] and n-gram[17] techniques, are computationally simple and can work well for literal overlap, but they lose effectiveness under strong paraphrase or translation obfuscation[18]. Amirzhanov et al.[18] note that this limitation has pushed the field away from symbolic and lexical strategies toward semantic representations better suited to multilingual comparison.

The introduction of multilingual neural models changed the research landscape. The first major multilingual transformer to gain widespread use was mBERT[19]. BERT itself introduced masked language modelling, in which some words are hidden and the model is trained to predict them from context. By training on very large corpora, the model learns rich contextual embeddings that capture meaning rather than simple word identity. XLM-R[20] extended this idea with a much larger training corpus, CC-100, and adopted RoBERTa[21]-style training improvements such as dynamic masking and larger batches. Newer families such as XLM-V[22], ModernBERT[23], and mmBERT[24] aim to improve multilingual coverage, tokenizer behaviour, or runtime efficiency. Recent studies report strong results for transformer-based multilingual embeddings in semantic comparison and paraphrase-oriented cross-lingual settings[25].

Commercial systems remain strong in monolingual academic workflows but weak in cross-lingual use cases. Turnitin[26] and iThenticate[27] are widely deployed because they can match suspicious passages against very large indexed repositories and generate practical reports for institutions. Regional platforms such as Odevzdej[28] and CRZP[29] play a similar role in Czech and Slovak environments.

Their main limitation is that they are designed primarily for detecting direct or near-direct overlap within a single language[18]. They are not built for multilingual environments and they function as plagiarism detectors rather than citation verification systems.

Citation verification is less mature than plagiarism detection, but several works are directly relevant to this thesis. Some research focuses on citation intent and rhetorical function, as in SciCite[30]. A much closer conceptual match is the study by Zeeb and Olsson[10], which explicitly asks whether large language models can evaluate citations in scholarly documents. They treat citation evaluation as a binary classification problem, using ChatGPT to assign a boolean result together with a short explanation[10]. Although the approach by Zeeb and Olsson was not designed as a practical system, it shows that LLMs can perform the core reasoning task of citation verification.

An even more direct systems-level point of comparison is Haan’s[31] SemanticCite, which proposes citation verification through AI-powered full-text analysis and evidence-based reasoning. SemanticCite is especially relevant because it moves beyond abstract discussion of citation quality and treats the task as a pipeline problem. The system must work with full-text documents, identify citations, retrieve and analyse source material, and produce evidence that can justify the verification result[31].

The need for citation-verification tools is not new. Wright and Armstrong[32] described citation accuracy as a structural weakness in scholarly communication. More recent studies are relevant here for three different reasons. First, Mogull[1] sharpens the definition of the problem by separating minor inaccuracies from major content errors, where the cited source fails to substantiate, is unrelated to, or contradicts the assertion. Second, Pavlovic et al.[2] provide empirical evidence that such claim-support problems do occur in published literature, finding at least one inaccurate citation in 11% and 15% of articles in their two datasets. Third, Mehregan[3] frames unsupported citation as an ethical issue as well, noting that some cases may reflect deliberate citation manipulation rather than simple carelessness. Taken together, these works justify treating citation verification as a distinct task concerned with evidential support, not only with reference formatting. Large language models are now also being tested in neighbouring tasks such as scientific-misinformation detection[33], but these approaches still raise questions of reproducibility, cost, and consistency.

Recent work also warns that threshold choices and dataset bias can distort reported performance, especially in imbalanced settings[34][35]. These observations

matter here because the evaluation of any proposed system will face similar trade-offs.

6 Solution Overview

The proposed system is meant to provide evidence for thesis reviewers during citation verification process. In practice, verifying a citation is often routine work: find the cited source, locate the relevant passage, and decide whether it really supports the claim made in the thesis. The goal of the solution is to automate these repetitive steps and return enough evidence for the evaluator to make the final decision.

For that reason, the problem is treated as a pipeline rather than as a single classifier. The system has to work first with a PDF document, then with bibliography metadata, and finally with short claim-evidence pairs. These are different kinds of data, so it makes more sense to handle them in separate stages.

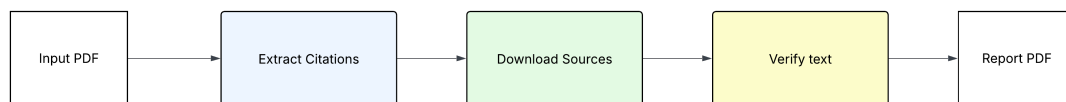


Figure 1: High-level pipeline of the proposed system

Figure 1 shows the overall logic. The input is an academic PDF document. The system extracts the citations actually used in the text, retrieves the referenced sources, compares the citing sentences with candidate passages from those sources, and produces a protocol-style PDF report. The output is therefore not just a verdict, but a trace of how the citation was checked.

6.1 Architecture and workflow

Figure 2 shows the detailed process of the pipeline. The system does not compare whole documents only. Citation misuse usually appears locally, at the level of a sentence or a short passage, so the comparison has to stay tied to individual citation contexts rather than to document-level similarity.

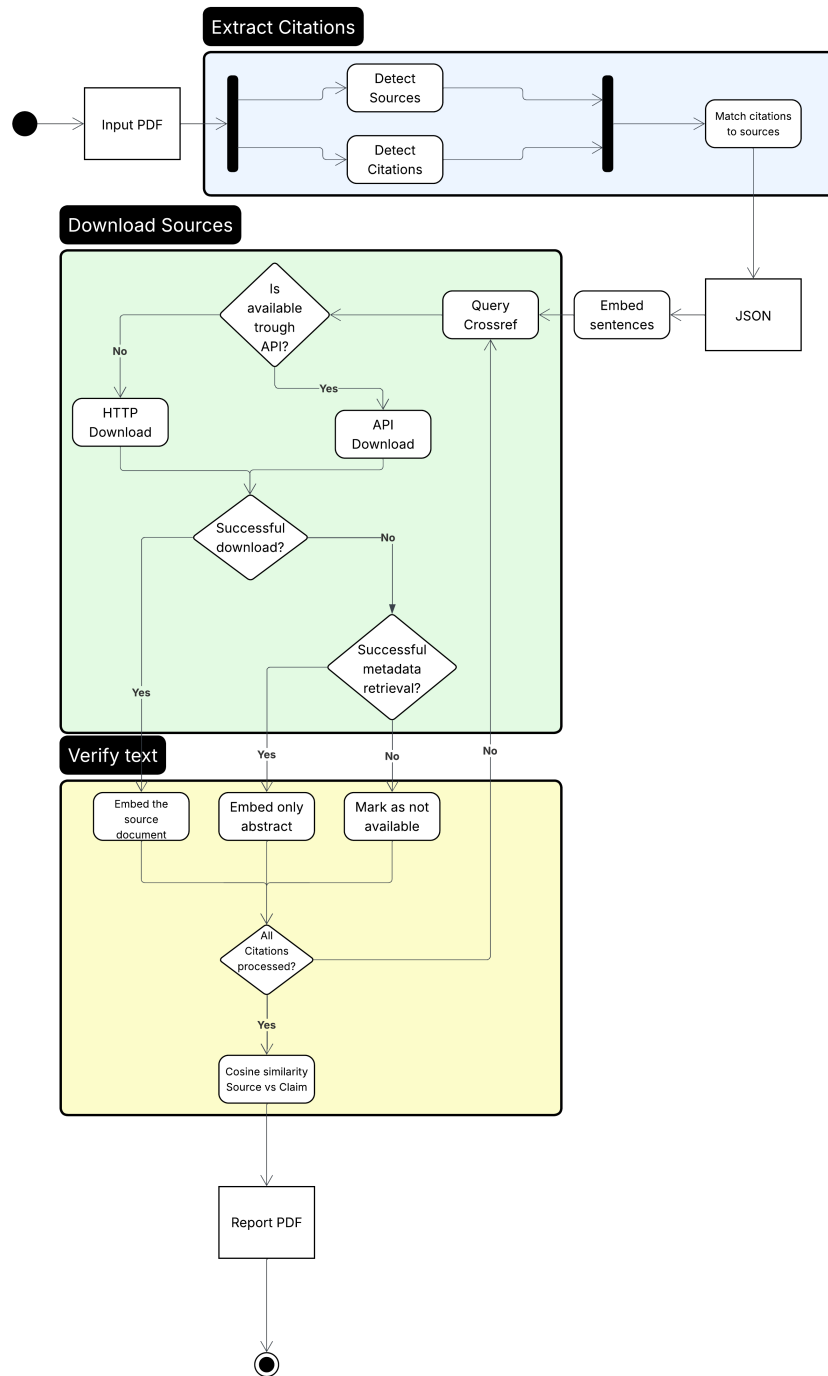


Figure 2: Detailed processing pipeline of the proposed system

6.1.1 Document understanding and reference alignment

The first stage processes the submitted PDF and extracts two layers in parallel: in-text citations and bibliography entries. These two outputs then have to be aligned, because later verification depends on linking each citation in the body text to a concrete item in the reference list. This citation-reference pair becomes the basic unit for the rest of the analysis.

The system is deliberately limited to IEEE-style numeric citations. Square-bracket references are easier to detect reliably using regular expressions than author-year formats, footnotes, or mixed institutional conventions, so this restriction keeps the focus on citation verification itself rather than on turning citation-style parsing into a separate problem.

After detection, each reference is converted into a structured internal representation containing the original reference string, available identifiers such as DOI, the citing sentences, and the number of occurrences in the text.

This stage also reveals one of the main limitations of the whole solution. PDF files are not designed for reliable machine interpretation. Bibliography sections may appear under different headings, line breaks can disturb sentence boundaries, and OCR noise can damage both citation detection and reference parsing. If the document is parsed incorrectly at this point, the later stages inherit that error.

6.1.2 Source retrieval and evidence preparation

Once the system identifies the references actually used in the text, it attempts to obtain the corresponding source documents. Retrieval follows a layered strategy. If a DOI is present in the reference, the system uses it in a Crossref API[36] request to obtain precise document metadata. If no DOI is available, it falls back to Crossref's full-text search. If neither approach returns a reliable match, the source is marked as unavailable.

When Crossref returns a probable match, the system then attempts to retrieve the document itself. It first checks the DOI and related metadata to determine whether the full text is accessible through one of the publisher APIs available to the system. This API step is important because many publishers protect their websites against automated scraping through bot detection, redirections, or session-based access controls, so direct HTTP requests are often less reliable than an official programmatic interface. If no API route is available, the system falls back to

standard HTTP requests. When the full text cannot be obtained, the system still tries to retrieve at least the abstract. If no usable source can be found, the reference remains marked as unavailable.

This is preferable to a single retrieval mechanism because academic sources vary widely in accessibility. Some are open, some expose only metadata, and others are blocked by publisher restrictions. Retrieval failure is therefore informative in its own right and should be reported explicitly rather than silently ignored.

After a source has been obtained, the text is segmented into smaller units (sentences, paragraphs, or sections) using GROBID[37]. This step is necessary because citation support tends to be localised. A paper may address the right topic in general while only specific passages directly support the claim attributed to it. Passage-level comparison is therefore more informative than whole-document similarity[38].

6.1.3 Semantic verification

The central analytical stage maps both the citing sentences and the retrieved source passages into a shared multilingual semantic space and saves the source passages into a vector database. Similarity is then computed between a claim in the thesis and candidate evidence passages using cosine similarity. This is suitable for cross-lingual verification because it does not rely on exact wording overlap and does not assume literal translation[25].

In the implemented design, this comparison is based on multilingual transformer embeddings, with XLM-R as the main model because it offers broad language coverage and solid performance in cross-lingual semantic tasks. The purpose of this stage is not to declare a citation true or false with certainty, but to rank the most relevant passages and show where support appears strongest.

6.2 Runtime architecture and external services

The pipeline does not run in isolation. Each stage interacts with a different set of external systems: bibliographic and publisher APIs during source retrieval, and a structured-extraction service together with a vector database during verification. Figure 3 captures this runtime view as a UML deployment diagram.

The application runs on a single host. GROBID and Milvus are deployed as local Docker containers. All other integrations are external HTTPS REST services, with optional embedding providers used only when configured.

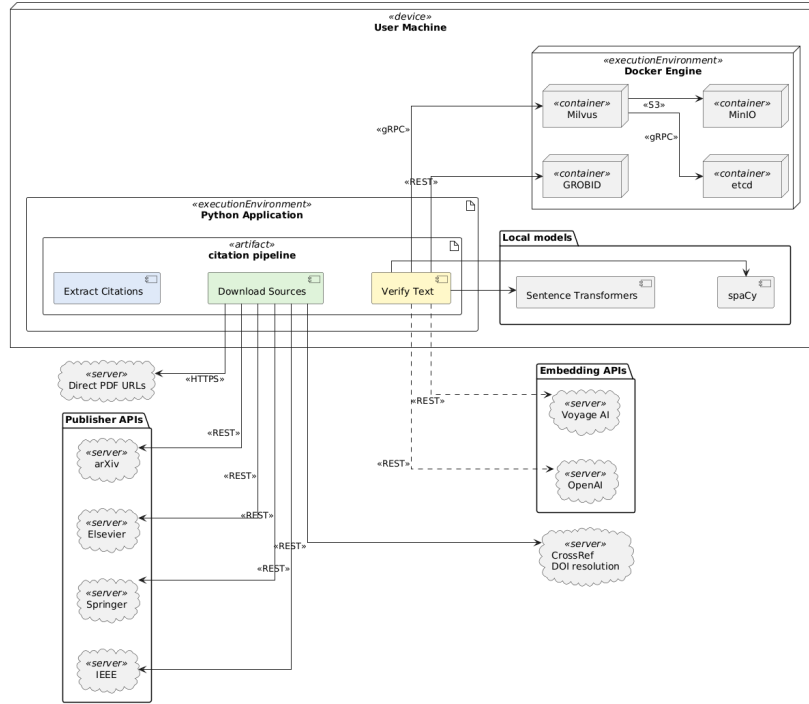


Figure 3: Runtime deployment and external service integrations

6.3 Reporting and limitations

The output of the pipeline is a protocol-style PDF report. It summarises the detected references, their frequency of use, unmatched citations, unavailable sources, and the passages that appear most relevant to each checked claim. This is more useful than a single numerical score because citation problems are heterogeneous and require context.

The main limitations remain straightforward. The system supports only one citation style, it depends heavily on PDF extraction quality and source availability, and its semantic component may still be weaker for Slovak or for highly specialised academic terminology than for high-resource English material. The support labels should therefore be understood as heuristic rather than definitive.

The system should not be understood as a fully automatic truth-assessment tool. High similarity suggests support, but it does not by itself prove correct interpretation. Low similarity is also not enough to prove misuse, because a valid citation may summarise a broader argument or depend on context outside a single sentence. The role of the system is therefore to narrow the search space for the human assessor, not to replace that assessor.

7 Risk Assessment

This project carries both technical and practical risks, but most of them can be reduced by keeping the scope narrow and by treating the system as decision support rather than as full automation.

Strengths. The main strength of the project is its focused scope. Instead of attempting to solve general multilingual plagiarism detection, the work concentrates on citation verification, which is easier to define, implement, and evaluate. A second strength is the modular pipeline design. Citation extraction, source retrieval, and semantic verification are separated, which makes errors easier to trace and the system easier to extend.

Weaknesses. The main weakness is the system’s strong dependence on input quality. PDF extraction can fail because of inconsistent formatting, OCR noise, or unusual bibliography structure. The solution is also limited to IEEE-style numeric citations, which improves reliability but reduces generality. Another weakness is its dependence on pretrained multilingual models, whose performance may be lower for low-resource languages or for highly specialised academic language[39]. Domain mismatch creates an additional problem because pretrained language models are often trained on general web data rather than academic prose[40]. Finally, the system does not produce a final verdict, only evidence, which may be less satisfying for users who want a clear true/false answer.

Opportunities. The project addresses a practical problem that is not well covered by standard plagiarism detection tools. If the core pipeline proves useful, it could be extended with a better interface, broader citation-style support, or deployment as an online service for schools. The same architecture could also be reused for other tasks in which claims need to be linked with source passages across documents.

Threats and mitigation. The most important external threat is source availability. Many cited works cannot be downloaded as full text because of publisher restrictions or broken links. This risk cannot be removed completely, so the mitigation strategy is to use layered retrieval, fall back to abstracts when possible, and report unavailable sources explicitly. Another threat is semantic uncertainty in cross-lingual comparison. Similarity scores may be affected by translation differences, paraphrasing, domain-specific terminology, or idiomatic expressions[41]. For that reason, the system is designed as a decision-support tool. It surfaces the most relevant evidence for reviewer assessment rather than replacing that assessment with a single automatic verdict. A final risk is scope growth. The topic could eas-

ily expand into citation parsing, web application development, or full plagiarism detection. This was mitigated by restricting the thesis to one citation style and one clearly defined verification workflow.

8 Experimental Reproducibility and Impact

8.1 Dataset and experimental scope

The main experiment processed a corpus of 1000 bachelor and master theses, with 250 documents each from four Computer Science and Software Engineering faculties across Slovak universities: Faculty of Informatics and Information Technologies STU Bratislava, Faculty of Electrical Engineering and Information Technology STU Bratislava, Faculty of Electrical Engineering and Information Technology UNIZA Žilina, and the Informatics programme at Faculty of Mathematics, Physics and Informatics UK Bratislava. All theses were submitted between 2020 and 2025 and downloaded from the Central Register of Theses and Dissertations (*CRZP*[29]). In the comparative figures, these are labelled Faculty 1 through Faculty 4 to avoid direct attribution of results to a specific institution.

Although the individual PDF files are publicly reachable through CRZP, the assembled dataset is not redistributed, as the students whose work was analysed did not give explicit consent for its use as research data. Researchers wishing to replicate the study can reconstruct an equivalent corpus by querying CRZP with the same programmes and year range.

8.2 Hardware and runtime environment

All pipeline stages were executed on the machine described in Table 1. The GPU was used exclusively by the local embedding model `paraphrase-xlm-r-multilingual-v1`[42] via `sentence-transformers`. All other stages ran on the CPU.

Table 1: Hardware used for the experiment

Component	Specification
CPU	AMD Ryzen 5 7600 (6-core, 12-thread)
GPU	NVIDIA GeForce RTX 4070 Ti Super (16 GB VRAM)
RAM	32 GB DDR5
OS	Windows 11

8.3 Reproducibility and open science

The pipeline and all analysis scripts are released publicly on GitHub. Dependencies are pinned in a `pyproject.toml` file and the supported Python version is explicitly stated in the project documentation. Two caching layers make results stable across runs: GROBID extractions are cached on disk keyed by DOI, and embedding vectors are stored in Milvus keyed by DOI and model name. Partial failures such as network timeouts can therefore be resumed without restarting from scratch.

The classification thresholds are explicit constants in the codebase: cosine similarity of at least 0.62 for *supported*, at least 0.42 for *related*, and below 0.42 for *no_support*. These should be treated as practical working thresholds rather than fully validated semantic boundaries. Ideally they would be tuned on an annotated dataset for the specific model in use. Both values remain configurable via command-line arguments so any downstream study can adjust them and regenerate all figures with the provided scripts.

Figure 4 illustrates why threshold choice is non-trivial. In a supplementary comparison over more than 200 cited sentences, the same citation-source pairs were embedded with three models: an XLM-R-based encoder, `voyage-3`, and `voyage-3-large`. The resulting score distributions differ clearly across all three, including between the two Voyage variants, confirming that thresholds are model-specific and should not be transferred between encoders without revalidation.

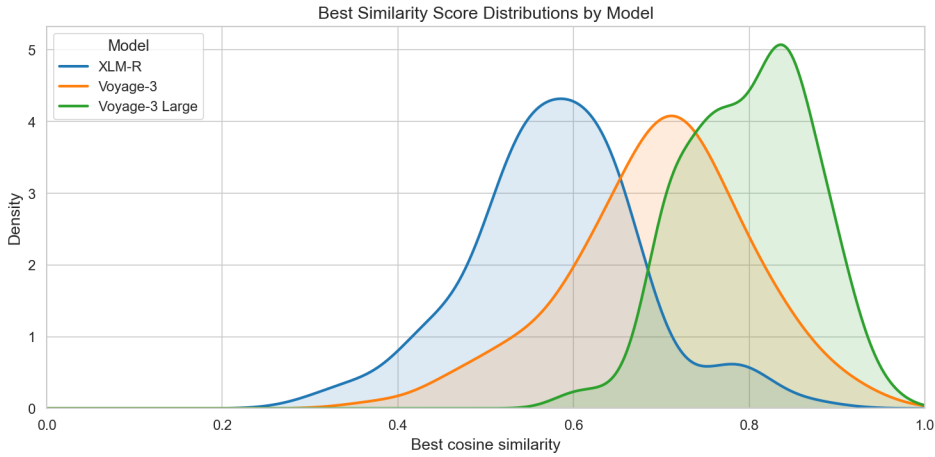


Figure 4: Distribution of best cosine similarity scores for the same cited sentence sample under three embedding models.

8.4 Reference corpus characteristics

Figure 5 shows the distribution of bibliography sizes across the four faculties. Most theses fall between 10 and 39 references, though the spread is broader in Faculties 2 and 3. Very large bibliographies are rare but do occur, with one thesis in Faculty 2 reaching the 180–189 bin. This variation matters because a short bibliography produces fewer citation-source pairs, while a long one places greater demand on retrieval.

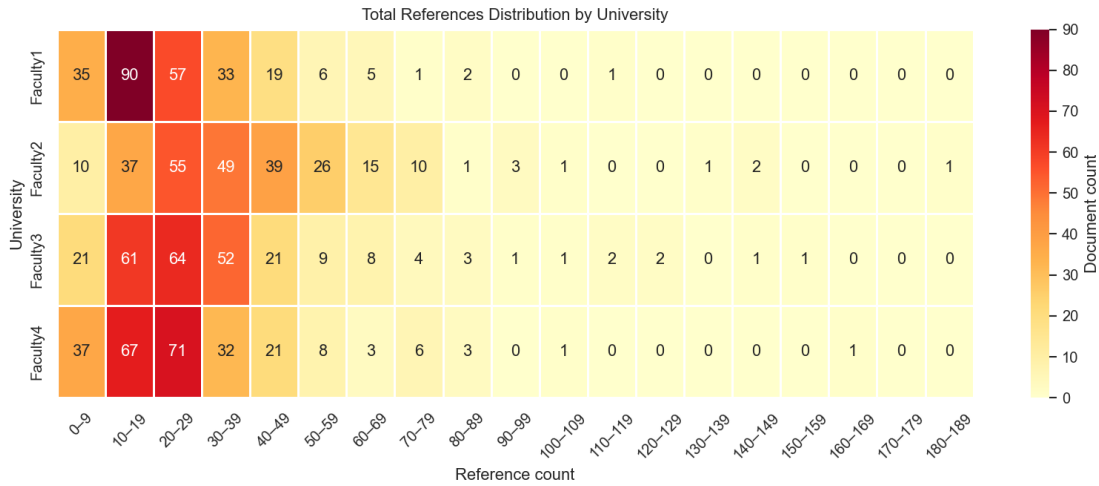


Figure 5: Distribution of bibliography sizes by faculty. Each cell shows the number of theses whose reference list falls within the corresponding bin.

Figure 6 shows how often individual references are cited within a thesis. The dominant pattern is single use in every faculty, ranging from 44.7% in Faculty 1 to 65.3% in Faculty 4, with repeated heavy reuse of four or more citations accounting for only 8.9%–17.9%. Most sources are therefore used locally for a limited number of claims.

Figure 7 shows mean DOI coverage per faculty. Faculty 1 has the lowest average at 42.8%, compared with 77.4% in Faculty 2, 60.1% in Faculty 3, and 70.2% in Faculty 4. The error bars reflect substantial within-faculty variation, meaning DOI coverage can differ considerably even between theses from the same faculty.

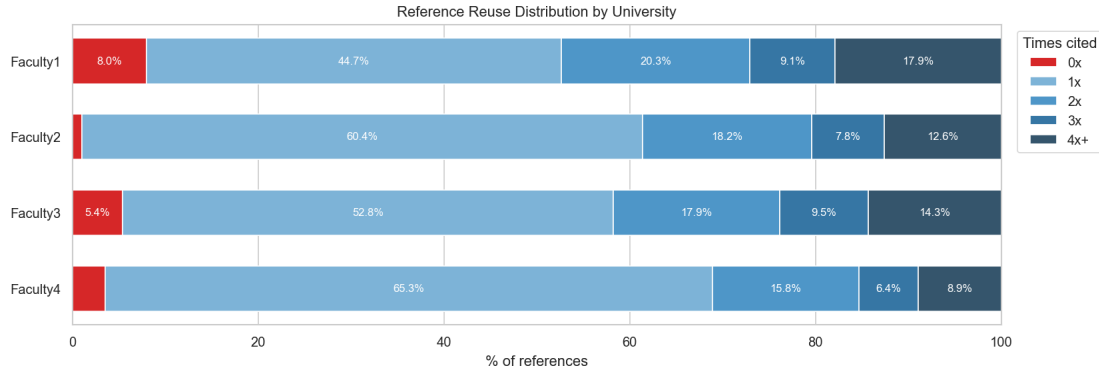


Figure 6: Reference reuse distribution by faculty. Each bar shows the proportion of cited references appearing once, twice, three times, or four or more times within a given thesis.

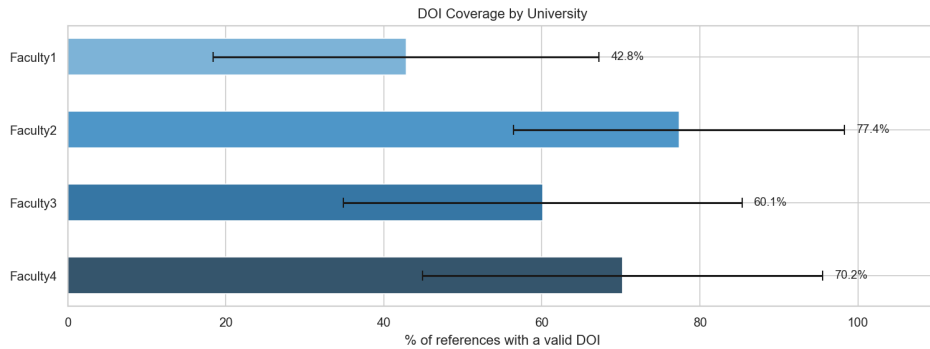


Figure 7: Mean DOI coverage by faculty.

8.5 Reference availability and retrieval results

The retrieval stage splits into two distinct problems: DOI-bearing references, which can often be resolved through metadata services and publisher APIs, and references without a DOI, which frequently point to web-native material. Figure 8 shows the resulting outcome breakdown across all four faculties.

Within the main run, downloadable full-text PDFs account for 14.5% of references in Faculty 1, 29.8% in Faculty 2, 21.9% in Faculty 3, and 28.2% in Faculty 4. Abstract-only fallback adds another 5.4%–10.4%. The *download_failed* category, ranging from 35.6% to 42.1% in Faculties 2–4, reflects both genuine retrieval failures and the limited publisher scope of this run. Citations whose source material was unavailable are excluded from the similarity analysis.

The main run used Crossref metadata resolution together with arXiv and Elsevier

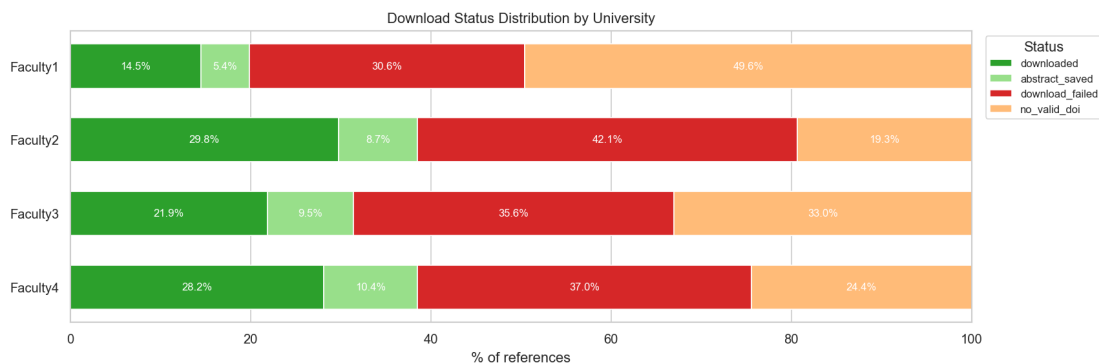


Figure 8: Source retrieval outcome distribution by faculty. Each bar shows the proportion of references falling into each retrieval category.

API access. IEEE and Springer integration was implemented but could not be applied across the full corpus before submission due to late access approval and restrictive API limits. Of the 20 631 DOI-bearing references, 7 175 full-text documents and 2 551 abstracts were successfully retrieved, corresponding to roughly 34% of the DOI-bearing subset excluding abstracts. A further 3 442 documents were potentially reachable via the IEEE API and 1 909 via Springer, suggesting the practical retrieval rate could have approached 60% with full publisher access.

Figure 9 shows why the decision was made to prioritize certain publishers. IEEE leads with 3 365 references, followed by Springer Nature, Elsevier, and arXiv. Beyond the top 15, the distribution is long-tailed with further publishers each contributing progressively smaller counts. A corpus from medicine, life sciences, or other disciplines would likely produce a different publisher ranking and a different optimal API mix.

For references without a DOI, Figure 10 shows the domain distribution among those that contained a URL. GitHub dominates with 292 references, reflecting the CS-oriented corpus, followed by Siemens, and Wikipedia. Youtube interestingly also occupies one of the top spots. An additional 2 751 no-DOI references contained no URL at all, so URL-based handling cannot close this gap entirely.

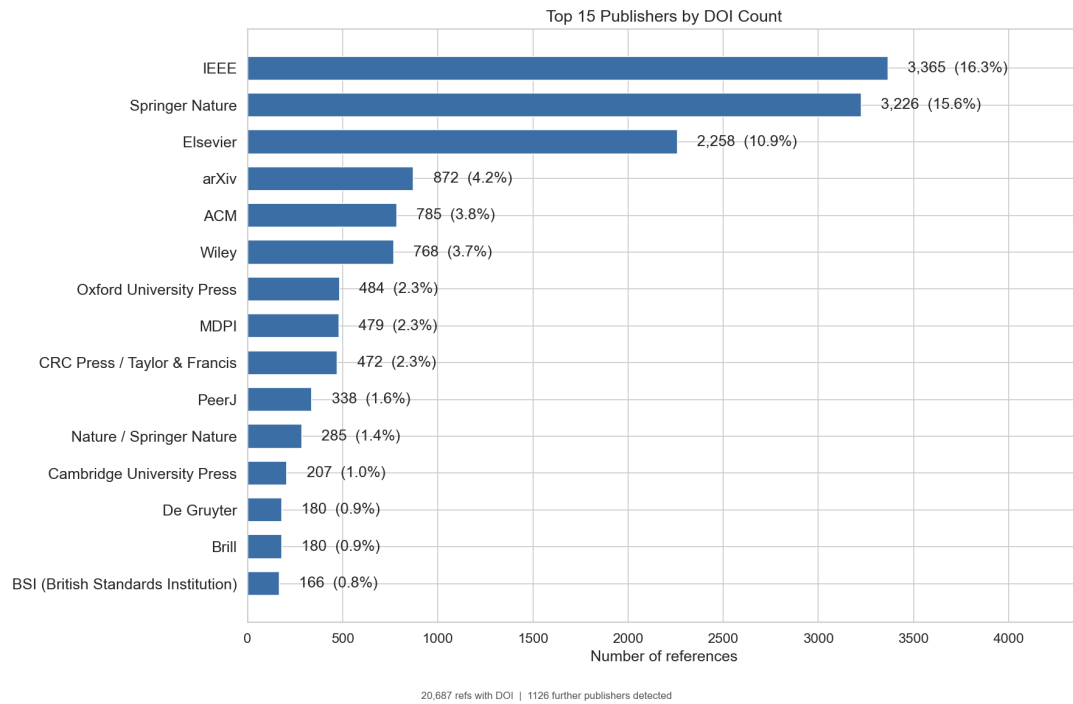


Figure 9: Top 15 publishers among references whose metadata contained a DOI.

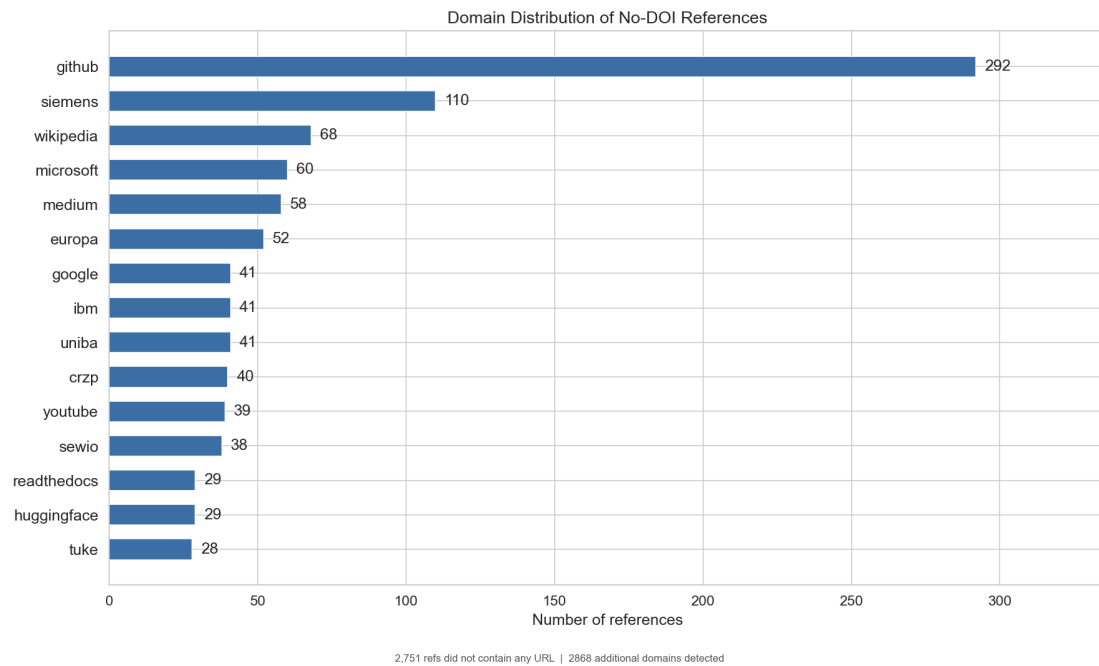


Figure 10: Top URL domains among references that did not contain a DOI.

8.6 Similarity analysis results

Figure 11 shows the overall distribution of best cosine similarity scores across all verified citations. The distribution is broad, concentrated roughly between 0.45 and 0.65, with very low and very high scores being less frequent. Verified citation-source pairs therefore do not split cleanly into two groups, which is consistent with citation use involving paraphrase, summary, and partial topical overlap rather than literal reuse.

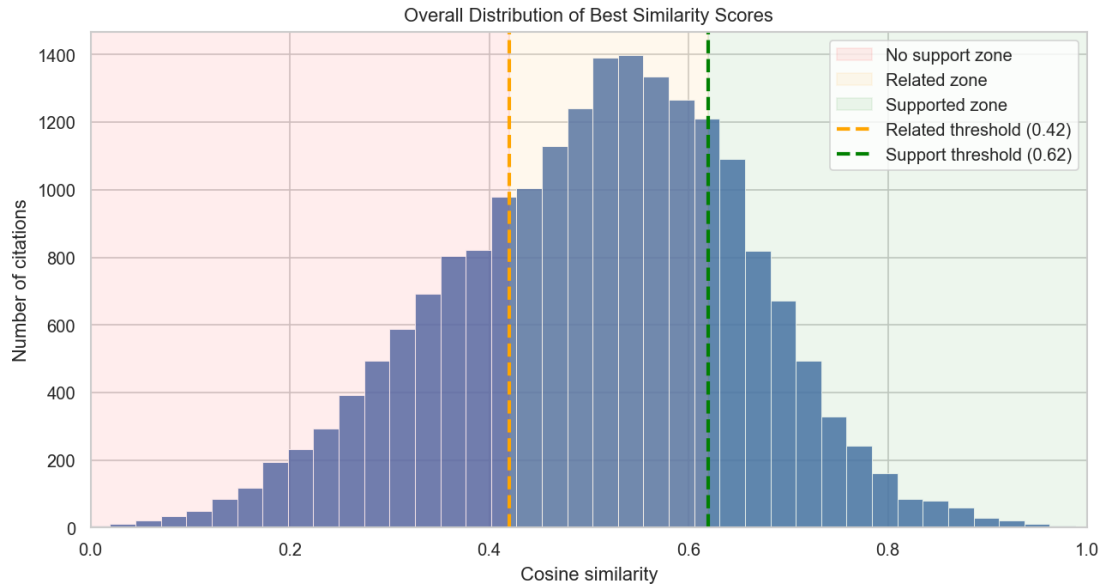


Figure 11: Distribution of best cosine similarity scores across all verified citations. Vertical dashed lines indicate the *related* and *supported* thresholds.

Figure 12 breaks the data down by faculty. The *related* label is the largest category everywhere, ranging from 43.3% in Faculty 1 to 50.7% in Faculty 3. The *supported* share ranges from 16.4% to 26.2% and *no_support* from 24.0% to 40.3%. Faculty 1 shows the lowest supported share and the highest no-support share, but this should be read cautiously given its weaker DOI coverage and lower retrieval success. Differences between faculties reflect retrieval constraints as much as citation practice, and no inference about institutional oversight is intended or supportable from these figures.

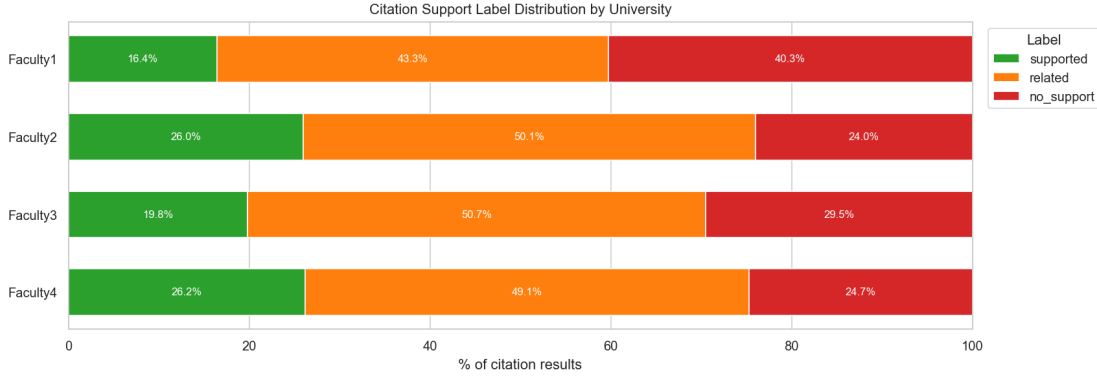


Figure 12: Citation support label distribution by faculty. Each bar shows the percentage of verified citations assigned the *supported*, *related*, or *no_support* label.

8.7 Interpretation and limitations

The results are indicative rather than conclusive. Two sources of imprecision accumulate across the pipeline. Citation and reference detection is not perfect: inline citations may be missed if the PDF has unusual formatting, and bibliography parsing can fail for entries deviating from IEEE style. Source retrieval was further constrained by both identifier coverage and publisher access scope, leaving many references excluded from verification entirely.

The most important interpretive caution concerns the *no_support* label. A low cosine similarity does not by itself indicate that a citation is misused. The student may have drawn a broader conclusion than any single passage captures, and paraphrase, cross-lingual reformulation, and high-level synthesis all tend to produce lower similarity scores than verbatim reuse. The pipeline’s role is to make verification more efficient for the reviewer: high-similarity cases can be cleared quickly, freeing time for lower-similarity citations that warrant closer inspection. Any case of genuine concern should be reviewed by a human assessor with access to both the thesis and the cited source before any conclusion is drawn.

A natural question is whether the label assignments should be validated against a human-annotated ground truth. Constructing such a dataset is non-trivial: annotation requires reading both the citing sentence and the source passage in depth, demands domain expertise, and is inherently subjective. Mogull[1] and Pavlovic et al.[2] operationalize citation inaccuracy differently even within the same research area, suggesting that inter-annotator agreement would itself be a significant challenge. A more appropriate evaluation for a decision-support tool would be a user study in which reviewers report whether the system reduced verification time and

directed attention to citations warranting closer inspection. This represents a natural direction for future work once the pipeline has been deployed in a real review setting.

9 Sustainability and Environmental Impact

The long-term value of this project depends on whether the solution can remain maintainable after the thesis is finished. Sustainability is supported by keeping the pipeline modular: citation extraction, source retrieval, and semantic verification are separated into stages that communicate through explicit intermediate outputs. A future maintainer can therefore modify one stage without redesigning the rest, and if a parsing tool, retrieval source, or embedding model becomes obsolete it can be replaced locally. External services such as GROBID, Crossref, and Milvus are accessed through well-defined interfaces, which lowers migration cost if one of them changes. The caching strategy described earlier also helps: once structured text or embeddings have been created, the pipeline reuses them in later runs instead of repeating expensive work or re-querying services that may become unavailable.

The system relies on a pretrained multilingual sentence embedding model rather than training from scratch, which significantly lowers the barrier for future use. New theses can be incorporated incrementally without retraining, and if a stronger encoder becomes available it can be swapped at the configuration level with only the similarity thresholds needing recalibration. By reusing pretrained models and caching intermediate results, the solution avoids the recurring energy cost of repeated fine-tuning. If the tool is later deployed for real institutional use, a centrally hosted service would be more sustainable than repeated local installations, making updates easier and limiting compute consumption to when verification is actually requested.

10 Employability

This project strengthened skills that are directly relevant to current IT and AI roles. It involved document parsing, information retrieval, multilingual NLP, vector search and usage of external APIs. These are practical tasks that appear in applied machine learning, search, and document-intelligence systems, so the thesis gave me experience that transfers well beyond the academic setting.

An important part of that experience came from working with Slovak and English texts in a setting where data quality and source availability were often imperfect. That forced me to think beyond ideal input and design a system that remains useful even when some parts fail or return uncertain results. At the same time, the project improved my ability to narrow a broad idea into a feasible scope, justify technical decisions, and explain the result clearly to other people. This combination of technical and communication skills strengthens my employability far more than solving an isolated programming task would.

11 Teamwork

Although the implementation was individual, the project developed through discussions with people from different backgrounds and through presenting the work at IITSRC. Those conversations were useful because they forced me to explain the solution in a practical way, not only as a technical pipeline. Instead of focusing only on models and similarity scores, I had to show what the tool would actually provide to a reviewer and where human judgment would still be necessary.

Teamwork was also important when defining the scope of the thesis. The original idea was a more general multilingual plagiarism detector, but during consultations it became clear that citation verification was a more focused and realistic direction. Presenting the work and discussing it with others helped confirm that this narrower scope had clearer evaluation criteria and more practical value for academic review. My role in that process was to act as the technical expert, propose implementable solutions, and translate feedback into concrete design decisions.

12 Conclusions

This thesis showed that cross-lingual citation verification can be implemented as a practical support tool for academic review. The result is not a system that replaces a human evaluator, but a pipeline that makes review work faster and more transparent by identifying citation contexts, retrieving relevant sources, and showing passages that appear to support the cited claim. From a technical point of view, the problem was addressed with a modular architecture built from existing tools and pretrained multilingual models, connecting PDF parsing, reference matching, source retrieval, passage comparison, and protocol generation into one process. The evaluation on 1000 Slovak bachelor and master theses confirmed that the approach is meaningful on a realistic corpus, though source availability emerged as

the main limiting factor — many references cannot be checked against full text even when the semantic model is usable.

The practical value of the project lies in supporting reviewers, librarians, and academic integrity staff with evidence rather than a single final score. As academic publishing becomes increasingly multilingual and cross-border, the ability to verify whether citations actually support the claims made around them becomes a shared challenge for research quality and integrity. A modular, open pipeline of this kind can be adapted by other institutions, integrated with existing repository infrastructure, and extended to other document types or citation styles without redesigning the core logic.

On a personal level, the project improved my ability to move from a broad idea to a narrower and more defensible technical solution. It required work across NLP, retrieval, software engineering, and evaluation, but also clear communication about uncertainty and practical limitations. That combination was one of the most valuable outcomes of the thesis.

The most natural future directions are improved usability and user-based evaluation. A web interface with reviewer feedback like showing detected citations and extracted sentences before verification begins, or allowing rejection or addition of retrieved texts. That would make the tool more suitable for institutional deployment, allow for more citation standards to be accepted, and provide a better retrieval rate. Quantitative tests are a useful first validation, but they do not show how helpful the tool feels in practice. Real reviewers reporting whether it made citation verification faster or clearer would provide a more informative picture.

13 Résumé

Táto bakalárska práca predstavuje návrh a experimentálne overenie systému na verifikáciu citácií v akademických textoch vo viacjazyčnom prostredí. Práca sa zameriava na prípady, keď autor uvedie citáciu na cudzojazyčný zdroj, no tvrdenie v texte týmto zdrojom nie je v skutočnosti podporené, hoci citácia môže na prvý pohľad pôsobiť formálne správne. Takéto situácie vznikajú najmä vtedy, keď autor obsah zdroja prekladá, parafrázuje, zjednodušuje alebo interpretuje nepresne. Cieľom navrhnutého riešenia nie je nahradiť odborný úsudok hodnotiteľa, ale zefektívniť proces kontroly citácií tým, že systém automaticky identifikuje citované tvrdenia, vyhledá relevantné zdroje a zobrazí pasáže, ktoré ich najpravdepodobnejšie podporujú.

Identifikácia problému

V akademickom prostredí sa správnosť citácií zvyčajne posudzuje ručne. Hodnotiteľ musí nájsť citovaný zdroj, lokalizovať príslušnú časť textu a posúdiť, či zdroj skutočne podporuje tvrdenie uvedené v práci. Tento proces je časovo náročný aj pri monolingválnych textoch a stáva sa ešte zložitejším v prípadoch, keď je citovaný zdroj napísaný v inom jazyku. Hlavný problém preto nespočíva v tom, že citácia chýba, ale v tom, že citovaný zdroj môže byť použitý nepresne, zavádzajúco alebo dokonca v rozpore s tvrdením, ktoré má podporovať.

Návrh riešenia

Navrhnuté riešenie pristupuje k problému ako k viackrokovému procesu. Systém spracúva vstupný PDF dokument, extrahuje citácie použité v texte a referencie zo zoznamu literatúry. Spája ich dokopy a následne sa pokúša získať zdroje citácií. Pri získavaní zdrojov využíva DOI identifikátory, službu Crossref a dostupné vydavateľské rozhrania API. Po získaní zdrojového textu systém rozdelí text na menšie pasáže a porovnáva ich s vetami obsahujúcimi citácie. Výstupom nie je len numerické skóre, ale protokolový report s prehľadom nájdených citácií, stavom získania zdrojov a najrelevantnejšími podpornými pasážami.

Implementácia

Architektúra riešenia pozostáva z troch hlavných častí: porozumenie dokumentu, získavanie zdrojov a sémantická verifikácia. V prvej fáze systém spracúva PDF dokument a extrahuje z neho numerické citácie numerického štandardu spolu s bibliografickými položkami. Z nich vytvára internú reprezentáciu obsahujúcu text referencie, prípadný DOI identifikátor, kontext citácie a počet výskytov v dokumente. V druhej fáze sa systém pokúša získať zdrojový text alebo aspoň abstrakt citovaného zdroja. Keďže akademické zdroje majú rôznu úroveň dostupnosti, riešenie explicitne rozlišuje medzi úspešne získanými dokumentmi, dokumentmi dostupnými len vo forme abstraktu a úplne nedostupnými zdrojmi.

Tretia fáza predstavuje jadro celého systému. Vety obsahujúce citácie a pasáže zo zdrojov sa mapujú do spoločného viacjazyčného sémantického priestoru pomocou transformerového embeddingového modelu. Ako hlavný model bol zvolený XLM-R, keďže poskytuje široké jazykové pokrytie a dobré vlastnosti pri porovnávaní významu medzi jazykmi. Podobnosť medzi tvrdením v práci a pasážou zo zdroja sa vyjadruje pomocou kosínovej podobnosti. Na základe dosiahnutého skóre systém zaraďuje výsledky do kategórií *supported*, *related* a *no_support*, pričom tieto štítky treba chápať ako heuristickú podporu pre hodnotiteľa, nie ako konečný automatický verdikt.

Experimenty a výsledky

Funkčnosť navrhnutého riešenia bola overovaná na korpuse 1000 bakalárskych a diplomových prác zo slovenských univerzít, konkrétne zo štyroch fakúlt zameraných na informatiku a softvérové inžinierstvo. V práci boli analyzované bibliografické charakteristiky korpusu, miera pokrytia DOI identifikátormi, úspešnosť získavania zdrojov a rozdelenie výsledných sémantických skóre. Experimenty ukázali, že veľká časť citácií spadá do stredného pásma podobnosti, čo zodpovedá reálnemu akademickému písaniu, kde sa zdroje často sumarizujú alebo parafrázujú namiesto doslovného preberania textu.

Výsledky zároveň potvrdili, že praktickým obmedzením celého riešenia nie je len presnosť samotného sémantického modelu, ale najmä kvalita vstupných PDF dokumentov, úspešnosť extrakcie bibliografie a dostupnosť zdrojových textov citovaných zdrojov. V experimentálnom behu bolo možné sémanticky verifikovať len tú časť citácií, pre ktoré sa podarilo získať dostatočne kvalitný zdrojový text alebo abstrakt. Získané výsledky preto nepredstavujú úplné hodnotenie kvality citovania v analyzovaných prácach, ale demonštrujú, že navrhnutý proces dokáže zmysluplne zúžiť priestor, ktorému sa má ľudský hodnotiteľ venovať podrobnejšie.

Prínosy, limity a budúci rozvoj

Hlavným prínosom práce je návrh uceleného pracovného postupu, ktorý prepája extrakciu citácií, získavanie zdrojov, viacjazyčné sémantické porovnávanie a generovanie dôkazovo orientovaného reportu do jedného procesu. Riešenie zároveň ukazuje, že verifikácia citácií je vhodnejšia ako podporný analytický nástroj než ako plne automatizovaný rozhodovací systém. Medzi hlavné limity patrí závislosť od kvality PDF extrakcie, obmedzenie na numerický citačný štýl a nižšia spoľahlivosť pri menej dostupných zdrojoch alebo pri špecializovanej terminológii.

Pri budúcom rozvoji by bolo vhodné rozšíriť riešenie o podporu ďalších citačných štýlov, zlepšiť používateľské rozhranie pre hodnotiteľov a doplniť možnosť manuálnej korekcie automaticky spárovaných citácií a zdrojov ešte pred samotnou verifikáciou. Praktický význam by mala aj používateľská evaluácia s reálnymi školiteľmi alebo knihovníkmi, ktorá by ukázala, do akej miery systém skutočne zrýchľuje a spresňuje kontrolu citácií v praxi. Navrhnutý systém tak vytvára základ pre ďalší vývoj nástrojov, ktoré by mohli doplniť existujúce antiplagiátorské riešenia o oblasť, ktorú dnes pokrývajú len obmedzene.

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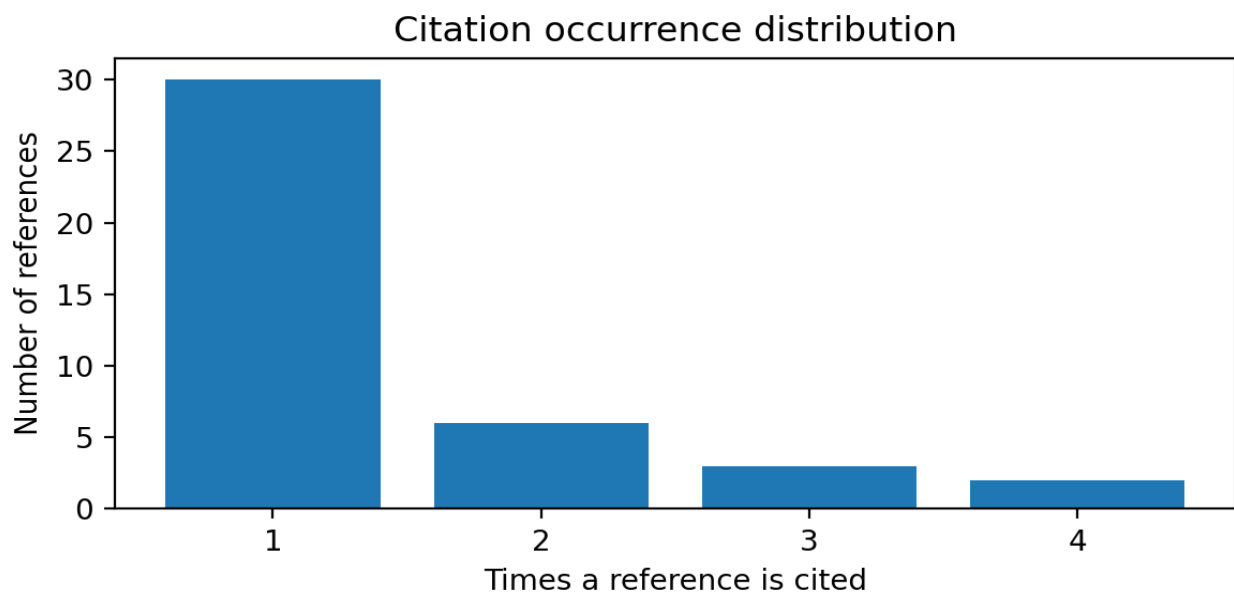
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Citation Verification Protocol

Source PDF: D:\Uni\bp-text\build\Bakalarka_Kubis.pdf

Paper statistics

Total references	41
Cited references	41
Unused references	0
Total citations (in-text)	60
Unmatched citation entries	0
Unmatched citation labels	0

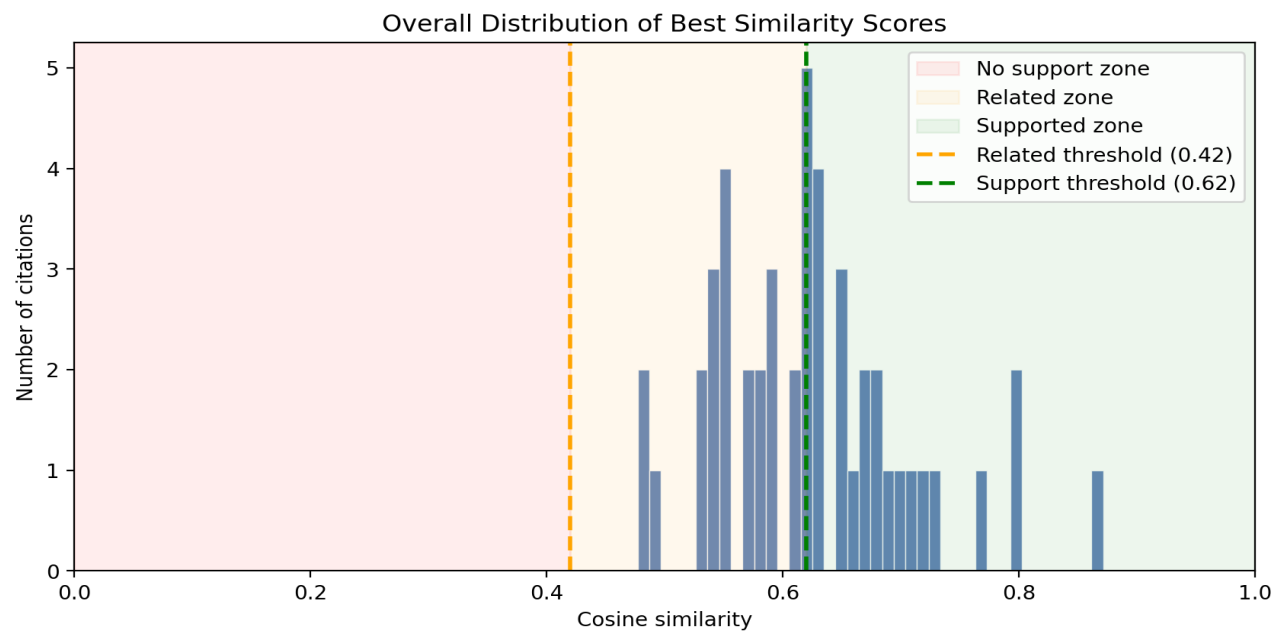


Summary

References (links): 41

Verified citing sentences: 47

Labels: supported=22, related=25, no_support=0



Reference [1]

Occurrences in work: 4

Resolved DOI: 10.1371/journal.pone.0184727

Resolved title: Accuracy of cited “facts” in medical research articles: A review of study methodology and recalculation of quotation error rate

Download status: already_indexed

Raw reference: [1] Scott A. Mogull. “Accuracy of cited “facts” in medical research articles: A review of study methodology and recalculation of quotation error rate”. In: PLOS ONE 12.9 (Sept. 2017), pp. 1–17. doi: 10.1371/journal.pone. 0184727. url: <https://doi.org/10.1371/journal.pone.0184727>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.688 **hint:** body **zone:** body

Citing: Recent studies show that this link between claim and source is not always reliable: a citation may be present, yet the cited work may only partially support the statement, be interpreted incorrectly, or be used in a misleading way[1][2][3].

- **Match 1** (sim 0.688) [body/sentence] Discussion: Alternatively, a reference may be inadvertently included in the work-cited list but not quoted in the text (as mentioned in the same study [26]).
- **Match 2** (sim 0.679) [body/sentence] Introduction: The primary sources cited may be misquoted, inapplicable, unreliable and occasionally even imaginary.
- **Match 3** (sim 0.669) [body/sentence] Discussion: Often, a reference listed in the work-cited list may be cited multiple times within an article and also might correspond to several different assertions (e.g., one study [26] mentioned an extreme case in which a single reference was associated with 13 in-text quotations).

2. Label: supported **sim:** 0.677 **hint:** body **zone:** body

Citing: Mogull distinguishes between minor inaccuracies and major content errors[1].

- **Match 1** (sim 0.677) [body/sentence] Discussion: g Includes secondary citations as minor content errors (the number of secondary citations was unable to be distinguished from the minor content errors).
- **Match 2** (sim 0.670) [body/sentence] Assigning consistent definitions and criteria for quotation errors: First, I defined “content errors” as an informative or “factual” inaccuracies and distinguished these errors from “source errors,” which are secondary (or indirect) citations.
- **Match 3** (sim 0.636) [body/paragraph] Assigning consistent definitions and criteria for quotation errors: Second, I applied consistent criteria for subcategorizing content errors into major and minor errors. The criteria for major and minor errors is more standardized throughout the original studies, although some minor differences existed. (The most notable difference in the criteria is that source errors are considered minor errors in approximately half of the studies.) As summarized in Table 2, a major content error was a cited assertion in which...

3. Label: supported **sim:** 0.660 **hint:** body **zone:** body

Citing: First, Mogull[1] sharpens the definition of the problem by separating minor inaccuracies from major content errors, where the cited source fails to substantiate, is unrelated to, or contradicts the assertion.

- **Match 1** (sim 0.660) [body/paragraph] Assigning consistent definitions and criteria for quotation errors: Second, I applied consistent criteria for subcategorizing content errors into major and minor errors. The criteria for major and minor errors is more standardized throughout the original studies, although some minor differences existed. (The most notable difference in the criteria is that source errors are considered minor errors in approximately half of the studies.) As summarized in Table 2, a major content error was a cited assertion in which...
- **Match 2** (sim 0.654) [body/sentence] Assigning consistent definitions and criteria for quotation errors: First, I defined “content errors” as an informative or “factual” inaccuracies and distinguished these errors from “source errors,” which are secondary (or indirect) citations.

- **Match 3** (sim 0.618) [body/sentence] Editor (emeritus): Such quotation errors undermine the scientific argument and foundation for the research being reported in the article and, moreover, may distort [2] and amplify [2][3][4] false information throughout the medical literature.

4. Label: related **sim:** 0.576 **hint:** unclear **zone:** body

Citing: Mogull[1] and Pavlovic et al.[2] operationalize citation inaccuracy differently even within the same research area, suggesting that inter-annotator agreement would itself be a significant challenge.

- **Match 1** (sim 0.576) [body/section] Editor (emeritus): The New England Journal of Medicine Since Ingelfinger's initial observations and commentary on the accuracy and reliability of quoted assertions in the medical literature, researchers have been systematically investigating the prevalence of inaccurately cited "facts" in published journal articles. Such quotation errors undermine the scientific argument and foundation for the research being reported in the article and, moreover, may distort [2] a...

- **Match 2** (sim 0.576) [body/paragraph] Editor (emeritus): The New England Journal of Medicine Since Ingelfinger's initial observations and commentary on the accuracy and reliability of quoted assertions in the medical literature, researchers have been systematically investigating the prevalence of inaccurately cited "facts" in published journal articles. Such quotation errors undermine the scientific argument and foundation for the research being reported in the article and, moreover, may distort [2] a...

- **Match 3** (sim 0.566) [body/sentence] Yes: In contrast to the strength of good to excellent inter-rater agreement for categorization of quotation errors, a resulting limitation was that each original study examined a single specialty corresponding to the expertise of the authors.

Reference [2]

Occurrences in work: 4

Resolved DOI: 10.1042/CS20201573

Resolved title: Literature cited

Download status: downloaded

Downloaded PDF: pdfs\nihms-1690179.pdf

Raw reference: [2] Vedrana Pavlovic et al. "How accurate are citations of frequently cited papers in biomedical literature?" In: Clin Sci (Lond) 135.5 (2021), pp. 671–681.

Citing sentences and best evidence

1. Label: supported **sim:** 0.634 **hint:** body **zone:** body

Citing: Recent studies show that this link between claim and source is not always reliable: a citation may be present, yet the cited work may only partially support the statement, be interpreted incorrectly, or be used in a misleading way[1][2][3].

- **Match 1** (sim 0.634) [body/sentence] Discussion: Although many studies have examined citation inaccuracies, our results may not be directly comparable due to differences in study design.

- **Match 2** (sim 0.632) [body/paragraph] Discussion: Although many studies have examined citation inaccuracies, our results may not be directly comparable due to differences in study design. Porrino et al. (11) used a similar approach to examine inaccuracies in citations of the Knirk and Jupiter (12) article, and found that 40% of citations were inaccurate by the time of the study (64/159). However, the generalizability of this finding was limited due to the fact that this study examined citations...

- **Match 3** (sim 0.628) [body/paragraph] Discussion: In contrast to our results, previous studies have reported associations between citation inaccuracies and citation style (Harvard vs. Vancouver) (1), the number of authors (one vs. more than one) (17) or the number of references (18). These divergent results may also be partially due to study design differences. In contrast to other studies (14,19,20), we have found an association between inaccurate citations and journal impact factor. There wer...

2. Label: related **sim:** 0.618 **hint:** unclear **zone:** body

Citing: Pavlovic et al.[2] use a similar support-based criterion, identifying error types such as nonexistent findings, incorrect interpretation of results, and chains of repeated citation errors.

- **Match 1** (sim 0.618) [body/paragraph] Introduction: While the academic literature is replete with similar errors, assessing citation accuracy takes considerable effort. Most scientists are aware of citation errors and copying, but few appreciate their prevalence or consequences. The purpose of this study was to: 1) examine how often frequently cited papers in biomedical literature are cited inaccurately, 2) explore factors associated with inaccurate citations and 3) discuss actions that authors, ...

- **Match 2** (sim 0.604) [body/sentence] Results: The most common finding was the citation of nonexistent findings (38.4%), followed by inaccurately cited numerical data/results (16.6%), inaccurate interpretation of findings (15.4%) and citations of quoted findings of another source (15.1%).

- **Match 3** (sim 0.602) [abstract/sentence] abstract: The most common problem was the citation of nonexistent findings (38.4%), followed by an incorrect interpretation of findings (15.4%).

3. Label: supported **sim:** 0.719 **hint:** body **zone:** abstract

Citing: Second, Pavlovic et al.[2] provide empirical evidence that such claim-support problems do occur in published literature, finding at least one inaccurate citation in 11% and 15% of articles in their two datasets.

- **Match 1** (sim 0.719) [abstract/sentence] abstract: At least one inaccurate citation was found in 11% and 15% of articles in the feasibility study and verification set, respectively, suggesting that inaccurate citations are common in biomedical literature.

- **Match 2** (sim 0.700) [body/section] •: In this study, we found that inaccurate citations are common in biomedical scientific literature. At least one inaccurate citation was found in 11% of the articles in feasibility study. This finding was confirmed in the verification set of articles, where citation inaccuracies were detected

in 15.0% of articles. The most common problem was the citation of nonexistent findings (38.4%), followed by an incorrect interpretation of findings (15.4%).

- **Match 3** (sim 0.700) [body/paragraph] •: In this study, we found that inaccurate citations are common in biomedical scientific literature. At least one inaccurate citation was found in 11% of the articles in feasibility study. This finding was confirmed in the verification set of articles, where citation inaccuracies were detected in 15.0% of articles. The most common problem was the citation of nonexistent findings (38.4%), followed by an incorrect interpretation of findings (15.4%).

4. Label: related **sim:** 0.555 **hint:** unclear **zone:** body

Citing: Mogull[1] and Pavlovic et al.[2] operationalize citation inaccuracy differently even within the same research area, suggesting that inter-annotator agreement would itself be a significant challenge.

- **Match 1** (sim 0.555) [body/sentence] Introduction: While the academic literature is replete with similar errors, assessing citation accuracy takes considerable effort.

- **Match 2** (sim 0.547) [body/paragraph] Introduction: While the academic literature is replete with similar errors, assessing citation accuracy takes considerable effort. Most scientists are aware of citation errors and copying, but few appreciate their prevalence or consequences. The purpose of this study was to: 1) examine how often frequently cited papers in biomedical literature are cited inaccurately, 2) explore factors associated with inaccurate citations and 3) discuss actions that authors, ...

- **Match 3** (sim 0.544) [body/section] •: Citation inaccuracies can undermine the integrity of the scientific literature. To reduce citation inaccuracies, several actions that authors, mentors and journals can take have been proposed.

Reference [3]

Occurrences in work: 3

Resolved DOI: 10.1177/17470161211068745

Resolved title: Scientific journals must be alert to potential manipulation in citations and referencing

Download status: already_indexed

Raw reference: [3] Mina Mehregan. "Scientific journals must be alert to potential manipulation in citations and referencing". In: Research Ethics 18.2 (2022), pp. 163–168. doi: 10.1177/17470161211068745. eprint: <https://doi.org/10.1177/17470161211068745>. url: <https://doi.org/10.1177/17470161211068745>.

Citing sentences and best evidence

1. Label: related **sim:** 0.619 **hint:** unclear **zone:** body

Citing: Recent studies show that this link between claim and source is not always reliable: a citation may be present, yet the cited work may only partially support the statement, be interpreted incorrectly, or be used in a misleading way[1][2][3].

- **Match 1** (sim 0.619) [body/sentence] Topic Piece: In other words, citation manipulation happens when there is no connection between the cited reference and the scholarly content of the manuscript, and the contained reference does not support the information provided in the manuscript (COPE Council, 2019; National Academy of Sciences, National Academy of Engineering, and Institute of Medicine, 2009).

- **Match 2** (sim 0.612) [body/sentence] Topic Piece: This may in part be attributable to unintentional errors in citation that are related to typographical mistakes and that can occur due to the negligence and carelessness of the authors; other times, however, it appears that there are many cases in which the errors in the references are due to academic misconduct (Wren and Georgescu, 2020), which is an unethical practice that involves knowingly and deliberately manipulating citations.

- **Match 3** (sim 0.540) [body/sentence] Topic Piece: After all, one of the simplest ways to inflate citation counts is to repeatedly self-cite or refer to the unrelated works of the authors' academic colleagues.

2. Label: related **sim:** 0.588 **hint:** unclear **zone:** body

Citing: Mehregan[3] further warns that some unsupported references Overview of Related Solutions may be intentional citation manipulation, used to increase citation counts rather than to support the argument.

- **Match 1** (sim 0.588) [body/sentence] Topic Piece: Citation manipulation is falsely citing a reference that was never utilized and is only included in the manuscript with the intention of increasing the number of its citations (COPE Council, 2019).

- **Match 2** (sim 0.563) [body/paragraph] Topic Piece: Citation manipulation for the purpose of increasing researchers' citation counts also happens when editors or peer reviewers of the manuscript ask researchers to add several unnecessary and unrelated references to the manuscript, which is called "coercive citation" and is unethical (Baas and Fennell, 2019; Van Noorden, 2020). Another dominant cause for this type of misconduct is that citation indexes play an important role in journals' rankings a...

- **Match 3** (sim 0.561) [body/sentence] Topic Piece: After all, one of the simplest ways to inflate citation counts is to repeatedly self-cite or refer to the unrelated works of the authors' academic colleagues.

3. Label: supported **sim:** 0.666 **hint:** body **zone:** body

Citing: Third, Mehregan[3] frames unsupported citation as an ethical issue as well, noting that some cases may reflect deliberate citation manipulation rather than simple carelessness.

- **Match 1** (sim 0.666) [body/sentence] Topic Piece: This may in part be attributable to unintentional errors in citation that are related to typographical mistakes and that can occur due to the negligence and carelessness of the authors; other times, however, it appears that there are many cases in which the errors in the references are due to academic misconduct (Wren and Georgescu, 2020), which is an unethical practice that involves knowingly and deliberately manipulating citations.

- **Match 2** (sim 0.543) [body/paragraph] Topic Piece: Unfortunately, the academic world is facing several types of scientific misconduct, such as plagiarism, data fabrication, and citation manipulation. Manipulating

citations is one of the forms of scientific misconduct with an alarming rate of increasing frequency (Ochsner Journal Blog, 2019) and hitherto has not received adequate attention from the scientific community. This may in part be attributable to unintentional errors in citation that are...

- **Match 3** (sim 0.529) [abstract/paragraph] abstract: Citation is an essential practice in scientific publishing. However, it is mandatory that citing the sources in a scientific work is performed in a proper manner. Manipulating citations in research articles is one form of academic research misconduct that violates publication ethics. Citation manipulation simply occurs for the purpose of increasing the number of citations of a researcher or a journal. Unfortunately, there has been a growing tren...

Reference [4]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [4] Zákon ■. 131/2002 Z. z. Zbierka zákonov Slovenskej republiky. 2002. url: <https://www.slov-lex.sk/ezbierky/pravne-predpisy/SK/ZZ/2002/131/20230101>.

Verification status: no_source_text

Reference [5]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [5] Zákon ■. 185/2015 Z. z. Zbierka zákonov Slovenskej republiky. 2015. url: <https://www.slov-lex.sk/ezbierky/pravne-predpisy/SK/ZZ/2015/185/>.

Verification status: no_source_text

Reference [6]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [6] Encyclopedia Britannica 2025. 2025. url: <https://www.britannica.com/topic/plagiarism>.

Verification status: no_source_text

Reference [7]

Occurrences in work: 1

Resolved DOI: 10.1109/incacct57535.2023.10141785

Resolved title: A Review on diverse algorithms used in the context of Plagiarism Detection

Download status: already_indexed

Raw reference: [7] Anchal Pokharana and Urvashi Garg. "A Review on diverse algorithms used in the context of Plagiarism Detection". In: 2023 International Conference on Advancement in Computation & Computer Technologies (InCACCT). 2023, pp. 1–6. doi: 10.1109/InCACCT57535.2023.10141785.

Citing sentences and best evidence

1. Label: supported **sim:** 0.675 **hint:** body **zone:** body

Citing: It also includes uncredited reuse of ideas, misleading paraphrase, false attribution, and other forms of source misuse discussed in the plagiarism-detection literature[7].

- **Match 1** (sim 0.675) [body/paragraph] • Problem of Protecting and Preserving Data: Along with that if you are rewriting or rephrased the preexisting work or thoughts into your own words, then this is also compulsory to cite the work, otherwise it is also the case of plagiarism. The Latin origins plagiarius, an enslaver, and plagiarism, to steal, are where the word plagiarism comes from. "The theft of another author's work and the presentation of it as one's own, constitutes plagiarism and is a significant violation of the ethi...

- **Match 2** (sim 0.668) [body/sentence] • Problem of Protecting and Preserving Data: Along with that if you are rewriting or rephrased the preexisting work or thoughts into your own words, then this is also compulsory to cite the work, otherwise it is also the case of plagiarism.

- **Match 3** (sim 0.663) [body/sentence] • Problem of Protecting and Preserving Data: Utilization of any pre-existing work or some preexisting words/statements in any form is considered at plagiarized content, according to the Oxford Dictionary.

Reference [8]

Occurrences in work: 1

Resolved DOI: 10.1016/j.urolonc.2010.09.013

Resolved title: The problem of plagiarism

Download status: already_indexed

Raw reference: [8] Melissa S. Anderson and Nicholas H. Steneck. "The problem of plagiarism". In: Urologic Oncology: Seminars and Original Investigations 29.1 (2011), pp. 90–94. issn: 1078-1439. doi: <https://doi.org/10.1016/j.urolonc.2010.09.013>. url: <https://www.sciencedirect.com/science/article/pii/S107814391000270X>.

Citing sentences and best evidence

1. Label: related **sim:** 0.535 **hint:** unclear **zone:** body

Citing: Anderson and Steneck[8] argue that the ethical problem does not arise from learning from prior work, but from presenting others' ideas or words as one's own without proper acknowledgment.

- **Match 1** (sim 0.535) [body/paragraph] Introduction: Researchers' careers are built on their ideas and their contributions to the research record. It should be no surprise, then, that plagiarism is viewed by the research community as a serious violation of the norms of research. It constitutes tampering with the system by which researchers' work is recognized and rewarded, and it is a personal affront and act of disrespect to the individual who wrote the original words or expressed the original id...

- **Match 2** (sim 0.534) [body/section] Introduction: Researchers' careers are built on their ideas and their contributions to the research record. It should be no surprise, then, that plagiarism is viewed by the research community as a serious violation of the norms of research. It constitutes tampering with the system by which researchers' work is recognized and rewarded, and it is a personal affront and act of disrespect to the individual who wrote the original words or expressed the original id...

- **Match 3** (sim 0.519) [body/sentence] Introduction: It constitutes tampering with the system by which researchers' work is recognized and rewarded, and it is a personal affront and act of disrespect to the individual who wrote the original words or expressed the original idea.

Reference [9]

Occurrences in work: 2

Resolved DOI: 10.1007/s10579-009-9114-z

Resolved title: Cross-language plagiarism detection

Download status: already_indexed

Raw reference: [9] Martin Potthast et al. "Cross-language plagiarism detection". In: Language Resources and Evaluation 45.1 (2011), pp. 45–62. issn: 1574-0218. doi: 10. 1007/s10579-009-9114-z. url: <https://doi.org/10.1007/s10579-0099114-z>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.710 **hint:** body **zone:** body

Citing: Cross-lingual plagiarism detection studies reuse across languages, including direct translation and paraphrased translation[9], while citation verification asks whether a cited source actually supports the statement to which it is attached[10].

- **Match 1** (sim 0.710) [body/paragraph] Introduction: The paper in hand investigates a particular kind of text plagiarism, namely the detection of plagiarism across languages, sometimes called translation plagiarism. The different kinds of text plagiarism are organized in Figure 1. Cross-language plagiarism, shown encircled, refers to cases where an author translates text from another language and then integrates the translated text into his/her own writing. It is reasonable to assume that plagiari...

- **Match 2** (sim 0.691) [body/sentence] Introduction: The paper in hand investigates a particular kind of text plagiarism, namely the detection of plagiarism across languages, sometimes called translation plagiarism.

- **Match 3** (sim 0.649) [body/sentence] Introduction: There are no studies which directly assess the amount of cross-language plagiarism, but in 2005 a broader study among 18 000 students revealed that almost 40% of them admittedly plagiarized at least once, which

2. Label: related **sim:** 0.574 **hint:** unclear **zone:** body

Citing: This kind of misuse is harder to detect than literal copying because the wording changes while the argumentative role of the source remains central[9].

- **Match 1** (sim 0.574) [body/sentence] Introduction: The manual analysis of text with respect to plagiarism becomes infeasible on a large scale, so that automatic plagiarism detection attracts considerable attention.

- **Match 2** (sim 0.567) [body/paragraph] Introduction: Plagiarism detection, the automatic identification of plagiarism and the retrieval of the original sources, is developed and investigated as a possible countermeasure. Although humans can identify cases of plagiarism in their areas of expertise quite easily, it requires much effort to be aware of all potential sources on a given topic and to provide strong evidence against an offender. The manual analysis of text with respect to plagiarism becom...

- **Match 3** (sim 0.555) [body/section] Large part of document Document model comparison Without reference: style analysis With reference: fingerprinting: Figure 1 Taxonomy of text plagiarism types, along with approaches to detect them [19]. also includes cross-lingual cases [16]. Apart from being an important practical problem, the detection of cross-language plagiarism also poses a research challenge, since the syntactical similarity between source sections and plagiarized sections found in the monolingual setting is more or less lost across languages. Hence, research on this task may help to im...

Reference [10]

Occurrences in work: 3

Resolved DOI: 10.1021/acs.chas.4c00097.s001

Resolved title: Can Large Language Models (LLMs) Act as Virtual Safety Officers?

Download status: already_indexed

Raw reference: [10] Ahmad Zeeb and Philip Olsson. Citation Evaluation Using Large Language Models (LLMs) : Can LLMs evaluate citations in scholarly documents? An experimental study on ChatGPT. 2024. References

Citing sentences and best evidence

1. Label: related **sim:** 0.591 **hint:** unclear **zone:** body

Citing: Cross-lingual plagiarism detection studies reuse across languages, including direct translation and paraphrased translation[9], while citation verification asks whether a cited source actually supports the statement to which it is attached[10].

- **Match 1** (sim 0.591) [body/sentence] Related work: Their research categorises citation errors into three categories: omissions, which occur when authors overlook relevant research, incorrect references, which involve errors in citing sources such as incorrect author names and wrong titles, and quotation errors, which misreport the findings of cited studies.
- **Match 2** (sim 0.584) [body/section] Related work: In 2008, Wright et al [14]. examined the prevalence and impact of faulty citations in academic literature. Their research categorises citation errors into three categories: omissions, which occur when authors overlook relevant research, incorrect references, which involve errors in citing sources such as incorrect author names and wrong titles, and quotation errors, which misreport the findings of cited studies. They highlight that a significant...
- **Match 3** (sim 0.584) [body/paragraph] Related work: In 2008, Wright et al [14]. examined the prevalence and impact of faulty citations in academic literature. Their research categorises citation errors into three categories: omissions, which occur when authors overlook relevant research, incorrect references, which involve errors in citing sources such as incorrect author names and wrong titles, and quotation errors, which misreport the findings of cited studies. They highlight that a significant...

2. Label: related **sim:** 0.616 **hint:** weak_abstract **zone:** abstract

Citing: A much closer conceptual match is the study by Zeeb and Olsson[10], which explicitly asks whether large language models can evaluate citations in scholarly documents.

- **Match 1** (sim 0.616) [abstract/sentence] abstract: This study investigates the capacity of Large Language Models (LLMs), specifically ChatGPT 3.5 and 4, to evaluate citations in scholarly papers.
- **Match 2** (sim 0.588) [abstract/paragraph] abstract: This study investigates the capacity of Large Language Models (LLMs), specifically ChatGPT 3.5 and 4, to evaluate citations in scholarly papers. Given the importance of accurate citations in academic writing, the goal was to determine how well these models can assist in verifying citations. A series of experiments were conducted using a dataset of our own creation. This dataset includes the three main citation categories: Direct Quotation, Parap...
- **Match 3** (sim 0.550) [abstract/sentence] abstract: Given the importance of accurate citations in academic writing, the goal was to determine how well these models can assist in verifying citations.

3. Label: supported **sim:** 0.665 **hint:** body **zone:** body

Citing: They treat citation evaluation as a binary classification problem, using ChatGPT to assign a boolean result together with a short explanation[10].

- **Match 1** (sim 0.665) [body/sentence] Model Evaluation: This classification task is treated as a binary classification problem.
- **Match 2** (sim 0.610) [body/sentence] Refine prompts:: This prompt will instruct ChatGPT to evaluate the validity of a citation and provide a boolean result along with a brief explanation.

- **Match 3** (sim 0.604) [body/section] Model Evaluation: To evaluate the models, a number of experiments is conducted, focusing on their ability to classify citations as valid or invalid. This classification task is treated as a binary classification problem. In classification problems, confusion matrices are often used as they summarise the classification performance of a classifier by comparing its predicted classifications to the actual truth [23]. In a binary classification problem, a confusion ma...

Reference [11]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.1801.06323

Resolved title: Plagiarism: Taxonomy, Tools and Detection Techniques

Download status: already_indexed

Raw reference: [11] Hussain A Chowdhury and Dhruba K Bhattacharyya. Plagiarism: Taxonomy, Tools and Detection Techniques. 2018. arXiv: 1801.06323 [cs.IR]. url: <https://arxiv.org/abs/1801.06323>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.801 **hint:** body **zone:** body

Citing: A widely cited taxonomy distinguishes deliberate copy-paste plagiarism, simple paraphrasing, patchwork paraphrasing, self-plagiarism, illegitimate-source plagiarism, retweet plagiarism, metaphor plagiarism, and idea plagiarism[11].

- **Match 1** (sim 0.801) [body/paragraph] Plagiarism and Its Types: Based on characteristics, plagiarism can also be categorized into literal and intelligent plagiarism. Literal plagiarism consists of copy-paste/clone, paraphrasing, self/recycled, and retweet plagiarism. The other form of plagiarism can be considered as intelligent type of plagiarism. In general, intrinsic plagiarism detection methods can detect paraphrasing, idea, and mosaic textual plagiarism sub-types [4], whereas, external plagiarism detecti...

- **Match 2** (sim 0.794) [body/sentence] Plagiarism and Its Types: Literal plagiarism consists of copy-paste/clone, paraphrasing, self/recycled, and retweet plagiarism.

- **Match 3** (sim 0.748) [body/sentence] Plagiarism and Its Types: In general, intrinsic plagiarism detection methods can detect paraphrasing, idea, and mosaic textual plagiarism sub-types [4], whereas, external plagiarism detection methods can detect clone, metaphor, retweet and possibly (with a low probability or none) self plagiarisms and error 404 [4].

Reference [12]

Occurrences in work: 2

Resolved DOI: 10.1016/j.ipm.2016.04.006

Resolved title: Candidate document retrieval for cross-lingual plagiarism detection using two-level proximity information

Download status: already_indexed

Raw reference: [12] Nava Ehsan and Azadeh Shakery. "Candidate document retrieval for crosslingual plagiarism detection using two-level proximity information". In: Information Processing & Management 52.6 (2016), pp. 1004–1017. issn: 03064573. doi: <https://doi.org/10.1016/j.ipm.2016.04.006>. url: <https://www.sciencedirect.com/science/article/pii/S0306457316300784>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.793 **hint:** body **zone:** body

Citing: Cross-lingual plagiarism itself is usually defined as reuse in which a suspicious document in one language originates from a document in another language, either through exact translation or paraphrased translation[12].

- **Match 1** (sim 0.793) [body/sentence] Related work: In cross-language plagiarism, the suspicious document in language L originates from another document in a language other than L.
- **Match 2** (sim 0.761) [body/sentence] Related work: In order to detect cross-lingual plagiarism, either cross-language similarity is used or the document is translated and monolingual similarity is used.
- **Match 3** (sim 0.709) [body/sentence] Introduction: In automatic cross-language plagiarism detection, the task is to retrieve plagiarized text written in language L that has originated from another document in a language other than L. With the rapid growth of documents in different languages, the increased accessibility of electronic documents, and the availability of translation tools, cross-language plagiarism has become a serious problem and its detection requires more attention.

2. Label: supported **sim:** 0.647 **hint:** body **zone:** body

Citing: Ehsan et al.[12] group the main comparison strategies into syntax based, dictionary based, parallel corpus based, comparable corpus based, and multilingual semantic network based approaches.

- **Match 1** (sim 0.647) [body/sentence] Related work: There are five kinds of cross-language similarity assessment models that have been proposed in the literature (Franco-Salvador, Gupta, & Rosso, 2013;Potthast et al., 2011), (1) syntax-based, (2) dictionary-based, (3) parallel corpus-based, (4) comparable corpus-based, and (5) multilingual semantic network based approaches.
- **Match 2** (sim 0.598) [body/paragraph] Related work: In order to detect cross-lingual plagiarism, either cross-language similarity is used or the document is translated and monolingual similarity is used. There are five kinds of cross-language similarity assessment models that have been proposed in the literature (Franco-Salvador, Gupta, & Rosso, 2013;Potthast et al., 2011), (1) syntax-based, (2) dictionary-based, (3) parallel corpus-based, (4) comparable corpus-based, and (5) multilingual semantic network based approaches.
- **Match 3** (sim 0.560) [body/sentence] Related work: -ASA (Barrón-Cedeño, Rosso, Pinto, & Juan, 2008), LSI (Dumais, Letsche, Littman, & Landauer, 1997) and KCCA (Vinokourov, Cristianini, & Shawe-taylor, 2002) use parallel corpora in order to find cross-language similarity, while Cross-Language Explicit Semantic Analysis (CL-ESA) (Potthast, Stein, & Anderka, 2008), which is the cross-lingual generalization of ESA (Gabrilovich & Markovitch, 2007), uses comparable corpora for this purpose.

Reference [13]

Occurrences in work: 2

Resolved DOI: 10.1145/3345317

Resolved title: Academic Plagiarism Detection

Download status: already_indexed

Raw reference: [13] Tomáš Foltýnek, Norman Meuschke, and Bela Gipp. “Academic Plagiarism Detection: A Systematic Literature Review”. In: ACM Comput. Surv. 52.6 (Oct. 2019). issn: 0360-0300. doi: 10.1145/3345317. url: <https://doi.org/10.1145/3345317>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.871 **hint:** body **zone:** body

Citing: Foltýnek et al.[13] note that “even in the best case, i.e., if the plagiarism is discovered, reviewing and punishing plagiarized research papers and grant applications still causes a high effort for the reviewers, affected institutions, and funding agencies”.

- **Match 1** (sim 0.871) [body/sentence] INTRODUCTION: Even in the best case, i.e., if the plagiarism is discovered, reviewing and punishing plagiarized research papers and grant applications still causes a high effort for the reviewers, affected institutions, and funding agencies.
- **Match 2** (sim 0.695) [body/paragraph] INTRODUCTION: Furthermore, academic plagiarism wastes resources. For example, Wager [261] quotes a journal editor stating that 10% of the papers submitted to the respective journal suffered from plagiarism of an unacceptable extent. In Germany, the ongoing crowdsourcing project VroniPlagfoot_0 has investigated more than 200 cases of alleged academic plagiarism (as of July 2019). Even in the best case, i.e., if the plagiarism is discovered, reviewing and punis...
- **Match 3** (sim 0.661) [body/paragraph] INTRODUCTION: If plagiarism remains undiscovered, then the negative effects are even more severe. Plagiarists can unduly receive research funds and career advancements as funding agencies may award grants for plagiarized ideas or accept plagiarized research papers as the outcomes of research projects. The artificial inflation of publication and citation counts through plagiarism can further aggravate the problem. Studies showed that some plagiarized papers ar...

2. Label: supported **sim:** 0.700 **hint:** body **zone:** body

Citing: If it remains undiscovered, plagiarists may gain academic credit, funding, or career advancement based on misappropriated work[13].

- **Match 1** (sim 0.700) [body/sentence] INTRODUCTION: Plagiarists can unduly receive research funds and career advancements as funding agencies may award grants for plagiarized ideas or accept plagiarized research papers as the outcomes of research projects.
- **Match 2** (sim 0.693) [body/paragraph] INTRODUCTION: If plagiarism remains undiscovered, then the negative effects are even more severe. Plagiarists can unduly receive research funds and career advancements as funding agencies may award grants for plagiarized ideas or accept plagiarized research papers as the outcomes of research projects. The artificial inflation of publication and citation counts through plagiarism can further aggravate the problem. Studies showed that some plagiarized papers ar...
- **Match 3** (sim 0.664) [body/sentence] INTRODUCTION: If plagiarism remains undiscovered, then the negative effects are even more severe.

Reference [14]

Occurrences in work: 1

Resolved DOI: 10.1177/0165551518787696

Resolved title: Cross-lingual text alignment for fine-grained plagiarism detection

Download status: already_indexed

Raw reference: [14] Nava Ehsan, Azadeh Shakery, and Frank Wm Tompa. "Cross-lingual text alignment for fine-grained plagiarism detection". In: J. Inf. Sci. 45.4 (Aug. 2019), 443–459. issn: 0165-5515. doi: 10.1177/0165551518787696. url: <https://doi.org/10.1177/0165551518787696>.

Citing sentences and best evidence

1. Label: related **sim:** 0.587 **hint:** weak_abstract **zone:** abstract

Citing: Early work relied on dictionary-based translations[14] and syntax-based approaches[15], but these tend to be computationally expensive and limited by available resources.

- **Match 1** (sim 0.587) [abstract/sentence] abstract: In both steps, our approach uses a dictionary to obtain translations of individual terms instead of using a machine translation system to convert longer passages from one language to another.
- **Match 2** (sim 0.560) [abstract/sentence] abstract: Fast and easy access to a wide range of documents in various languages, in conjunction with the wide availability of translation and editing tools, has led to the need to develop effective tools for detecting cross-lingual plagiarism.
- **Match 3** (sim 0.439) [abstract/paragraph] abstract: Fast and easy access to a wide range of documents in various languages, in conjunction with the wide availability of translation and editing tools, has led to the need to develop effective tools for detecting cross-lingual plagiarism. Given a suspicious document, cross-lingual plagiarism detection comprises two main subtasks: retrieving documents that are candidate sources for that document and analysing those candidates one by one to determine ...

Reference [15]

Occurrences in work: 1

Resolved DOI: 10.1016/j.eswa.2017.07.006

Resolved title: Text plagiarism classification using syntax based linguistic features

Download status: already_indexed

Raw reference: [15] Vani K and Deepa Gupta. "Text plagiarism classification using syntax based linguistic features". In: Expert Systems with Applications 88 (2017), pp. 448– 464. issn: 0957-4174. doi: <https://doi.org/10.1016/j.eswa.2017.07.006>. url: <https://www.sciencedirect.com/science/article/pii/S095741741730475X>.

Citing sentences and best evidence

1. Label: related **sim:** 0.493 **hint:** unclear **zone:** body

Citing: Early work relied on dictionary-based translations[14] and syntax-based approaches[15], but these tend to be computationally expensive and limited by available resources.

- **Match 1** (sim 0.493) [body/paragraph] Conclusions and future work: In the future, exploration of machine learning approaches can to be extended to exhaustive levels of plagiarism detection. Unavailability of the larger data bases with manual plagiarism cases is another problem and hence the creation of such data bases are also required for the proper testing of the detection approaches. Further, structural and citation information have to be explored when dealing with scientific publications, which is a challen...

- **Match 2** (sim 0.446) [body/sentence] Existing systems: In both the existing systems the utilization of NLP techniques was not done, except for lexical generalization, which was utilized by Chong et al.

- **Match 3** (sim 0.440) [body/sentence] Existing systems: Thus, some of the reported systems have utilized syntactic linguistic information for source retrieval tasks.

Reference [16]

Occurrences in work: 1

Resolved DOI: 10.1109/conmedia46929.2019.8981821

Resolved title: String Matching based Plagiarism Detection for Document in Bahasa Indonesia

Download status: downloaded

Downloaded PDF:

pdfs\String_Matching_based_Plagiarism_Detection_for_Document_in_Bahasa_Indonesia.pdf

Raw reference: [16] Wayan Gede Suka Parwita, I Gusti Ayu Agung Diatri Indradewi, and I Nyoman Saputra Wahyu Wijaya. "String Matching based Plagiarism Detection for Document in Bahasa Indonesia". In: 2019 5th International Conference on New Media Studies (CONMEDIA). 2019, pp. 54–58. doi: 10.1109/CONMEDIA46929.2019.8981821.

Verification status: empty_reference_index

Reference [17]

Occurrences in work: 1

Resolved DOI: 10.3115/v1/d14-1153

Resolved title: Intrinsic Plagiarism Detection using N-gram Classes

Download status: already_indexed

Raw reference: [17] Imene Bensalem, Paolo Rosso, and Salim Chikhi. "Intrinsic Plagiarism Detection using N-gram Classes". In: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP). Ed. by Alessandro Moschitti, Bo Pang, and Walter Daelemans. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1459–1464. doi: 10.3115/v1/ D14-1153. url: <https://aclanthology.org/D14-1153/>.

Citing sentences and best evidence

1. Label: related **sim:** 0.529 **hint:** unclear **zone:** body

Citing: Classical retrieval-oriented or lexical methods, including string matching[16] and n-gram[17] techniques, are computationally simple and can work well for literal overlap, but they lose effectiveness under strong paraphrase or translation obfuscation[18].

- **Match 1** (sim 0.529) [body/paragraph] Introduction and Related Works: As opposed to the problem of authorship clustering, where the task is merely to attribute already defined fragments of a given document to different authors, the segmentation is a crucial and inevitable task in a real scenario of intrinsic plagiarism detection. Indeed, a granular segmentation may lead to an undependable style analysis, and a coarse segmentation may prevent the identification of the short plagiarized texts.

- **Match 2** (sim 0.522) [body/sentence] Introduction and Related Works: Indeed, a granular segmentation may lead to an undependable style analysis, and a coarse segmentation may prevent the identification of the short plagiarized texts.

- **Match 3** (sim 0.490) [body/paragraph] Introduction and Related Works: Unlike the authorship attribution and verification, where the examined text and the potential author text are separate (and hence their writing styles could be readily characterized and compared), these two parts are both merged in the same document with unknown boundaries. Furthermore, the plagiarized fragments in a suspicious document might stem from different authors, which renders the computational characterization of plagiarism difficult.

Reference [18]

Occurrences in work: 3

Resolved DOI: 10.3389/fcomp.2025.1504725

Resolved title: Plagiarism types and detection methods: a systematic survey of algorithms in text analysis

Download status: already_indexed

Raw reference: [18] Altynbek Amirzhanov, Cemil Turan, and Alfira Makhmutova. "Plagiarism types and detection methods: a systematic survey of algorithms in text analysis". In: *Frontiers in Computer Science* Volume 7 - 2025 (2025). issn: 2624-9898. doi: 10 . 3389 / fcomp . 2025 . 1504725. url: <https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2025.1504725>. References

Citing sentences and best evidence

1. Label: supported **sim:** 0.729 **hint:** body **zone:** body

Citing: Classical retrieval-oriented or lexical methods, including string matching[16] and n-gram[17] techniques, are computationally simple and can work well for literal overlap, but they lose effectiveness under strong paraphrase or translation obfuscation[18].

- **Match 1** (sim 0.729) [body/sentence] . Trade-offs between computational efficiency and detection accuracy: methods such as n-grams and string matching are computationally efficient but struggle with detecting paraphrased plagiarism.

- **Match 2** (sim 0.691) [body/sentence] Computationally efficient but less accurate methods: • Textual and lexical approaches: traditional methods like string matching and n-gram models excel in computational efficiency due to their simplicity and low resource requirements.

- **Match 3** (sim 0.677) [body/section] Computationally efficient but less accurate methods: • Textual and lexical approaches: traditional methods like string matching and n-gram models excel in computational efficiency due to their simplicity and low resource requirements. For example, Liu et al. (2015) demonstrated a linear complexity algorithm for source code detection with a 0% false alarm rate. However, these methods struggle with paraphrased or obfuscated plagiarism, limiting their robustness. • Citation-based approaches: techniqu...

2. Label: related **sim:** 0.479 **hint:** unclear **zone:** body

Citing: Amirzhanov et al.[18] note that this limitation has pushed the field away from symbolic and lexical strategies toward semantic representations better suited to multilingual comparison.

- **Match 1** (sim 0.479) [body/sentence] Conclusion: With continued advancements in AI and linguistic analysis, the field is well-positioned to develop more sophisticated, adaptable, and ethical solutions to combat plagiarism in an increasingly digital world.

- **Match 2** (sim 0.452) [body/paragraph] . . Concept-based approaches: Concept-based methods use semantic role labeling (SRL), named entity recognition (NER), and linguistic knowledge to detect idea plagiarism (Taufiq et al., 2023; Vani and Gupta, 2017b; Abdi et al., 2015). These approaches combine semantic and syntactic similarity to detect deeper textual relationships. A common metric used is Wu-Palmer Similarity:

- **Match 3** (sim 0.448) [body/paragraph] . Types of plagiarism: • Obfuscated plagiarism: modifying text structure or replacing key terms while retaining the original meaning (Alzahrani et al., 2015; Gharavi et al., 2019). • Cross-language plagiarism: translating content across languages without credit, making detection more complex (Alzahrani and Aljuaid, 2022; Franco-Salvador et al., 2016a). • Multilingual and language-independent plagiarism: extending plagiarism detection across different languages and lingu...

3. Label: related **sim:** 0.553 **hint:** unclear **zone:** body

Citing: Overview of Related Solutions Their main limitation is that they are designed primarily for detecting direct or near-direct overlap within a single language[18].

- **Match 1** (sim 0.553) [body/paragraph] . Textual similarity-based approaches: Textual similarity-based methods focus on detecting overlaps in surface-level textual features. These methods include:

- **Match 2** (sim 0.534) [body/sentence] . Textual similarity-based approaches: Textual similarity-based methods focus on detecting overlaps in surface-level textual features.

- **Match 3** (sim 0.522) [body/section] . Semantic similarity-based approaches: Semantic similarity methods detect plagiarism by analyzing the meaning behind words, going beyond surface-level text comparison. These methods are crucial for detecting paraphrased content and are divided into concept-based approaches and word embedding models.

Reference [19]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/N19-1423.url:<https://aclanthology.org/N19-1423/>

Download status: no_valid_doi

Raw reference: [19] Jacob Devlin et al. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Ed. by Jill Burstein, Christy Doran, and Thamar Solorio. Minneapolis, Minnesota, June 2019. doi: 10.18653/v1/ N19-1423. url: <https://aclanthology.org/N19-1423/>.

Verification status: no_source_text

Reference [20]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/2020.acl-main.747

Resolved title: Unsupervised Cross-lingual Representation Learning at Scale

Download status: already_indexed

Raw reference: [20] Alexis Conneau et al. “Unsupervised Cross-lingual Representation Learning at Scale”. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. Ed. by Dan Jurafsky et al. Online: Association for Computational Linguistics, July 2020, pp. 8440–8451. doi: 10.18653/v1/2020.acl-main.747. url: <https://aclanthology.org/2020.aclmain.747/>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.650 **hint:** body **zone:** body

Citing: XLM-R[20] extended this idea with a much larger training corpus, CC-100, and adopted RoBERTa[21]-style training improvements such as dynamic masking and larger batches.

- **Match 1** (sim 0.650) [body/sentence] Related Work: Inspired by RoBERTa, we show that mBERT and XLM are undertuned, and that simple improvements in the learning procedure of unsupervised MLM leads to much better performance.
- **Match 2** (sim 0.585) [body/sentence] Model: While XLM-R is among the largest models partly due to its large embedding layer, it has a similar number of parameters than XLM-100, and remains significantly smaller than recently introduced Transformer models for multilingual MT and transfer learning.
- **Match 3** (sim 0.577) [body/section] Introduction: The goal of this paper is to improve cross-lingual language understanding (XLU), by carefully studying the effects of training unsupervised crosslingual representations at a very large scale. We present XLM-R a transformer-based multilingual masked language model pre-trained on text in 100 languages, which obtains state-of-the-art performance on cross-lingual classification, sequence labeling and question answering. Multilingual masked language ...

Reference [21]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.1907.11692

Resolved title: RoBERTa: A Robustly Optimized BERT Pretraining Approach

Download status: already_indexed

Raw reference: [21] Yinhan Liu et al. RoBERTa: A Robustly Optimized BERT Pretraining Approach. 2019. arXiv: 1907.11692 [cs.CL]. url: <https://arxiv.org/abs/1907.11692>.

Citing sentences and best evidence

1. Label: related **sim:** 0.609 **hint:** unclear **zone:** body

Citing: XLM-R[20] extended this idea with a much larger training corpus, CC-100, and adopted RoBERTa[21]-style training improvements such as dynamic masking and larger batches.

- **Match 1** (sim 0.609) [body/sentence] Related Work: Our improved pretraining procedure, which we call RoBERTa, achieves state-of-the-art results on GLUE, RACE and SQuAD, without multi-task finetuning for GLUE or additional data for SQuAD.
- **Match 2** (sim 0.608) [body/sentence] Results: 9 Finally, we pretrain RoBERTa for significantly longer, increasing the number of pretraining steps from 100K to 300K, and then further to 500K.
- **Match 3** (sim 0.599) [body/paragraph] Introduction: We present a replication study of BERT pretraining (Devlin et al., 2019), which includes a careful evaluation of the effects of hyperparameter tuning and training set size. We find that BERT was significantly undertrained and propose an improved recipe for training BERT models, which we call RoBERTa, that can match or exceed the performance of all of the post-BERT methods. Our modifications are simple, they include: (1) training the model longer,...

Reference [22]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/2023.emnlp-main.813

Resolved title: XLM-V: Overcoming the Vocabulary Bottleneck in Multilingual Masked Language Models

Download status: already_indexed

Raw reference: [22] Davis Liang et al. XLM-V: Overcoming the Vocabulary Bottleneck in Multilingual Masked Language Models. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore, Dec. 2023. doi: 10.18653/v1/2023.emnlp-main. 813. url: <https://aclanthology.org/2023.emnlp-main.813/>.

Citing sentences and best evidence

1. Label: related **sim:** 0.543 **hint:** unclear **zone:** body

Citing: Newer families such as XLM-V[22], ModernBERT[23], and mmBERT[24] aim to improve multilingual coverage, tokenizer behaviour, or runtime efficiency.

- **Match 1** (sim 0.543) [body/sentence] Building larger vocabularies: Our work seeks to combine and improve upon existing methods for building large-scale vocabularies, pretrain with substantially bigger datasets, and explore vocabularies of 1M tokens and beyond.

- **Match 2** (sim 0.542) [abstract/sentence] abstract: Tokenizations using our vocabulary are typically more semantically meaningful and shorter compared to XLM-R. Leveraging this improved vocabulary, we train XLM-V, a multilingual language model with a one million token vocabulary.

- **Match 3** (sim 0.537) [body/sentence] Sentencepiece: This method is used by both XLM-R and our work. et al. (2020) proposed an approach to multilingual vocabulary construction that balances the trade-off between optimizing for cross-lingual subword sharing and the need for robust representation of individual languages.

Reference [23]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2412.13663

Resolved title: Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference

Download status: already_indexed

Raw reference: [23] Benjamin Warner et al. "Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference". In: (2024). arXiv: 2412.13663 [cs.CL]. url: <https://arxiv.org/abs/2412.13663>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.645 **hint:** body **zone:** body

Citing: Newer families such as XLM-V[22], ModernBERT[23], and mmBERT[24] aim to improve multilingual coverage, tokenizer behaviour, or runtime efficiency.

- **Match 1** (sim 0.645) [body/paragraph] Introduction: Recent modernization efforts have only partially addressed the shortcomings of encoder-only mod-els due to limited breadth. MosaicBERT (Portes et al., 2023), CrammingBERT (Geiping and Goldstein, 2023), and AcademicBERT (Izsak et al., 2021) focused on matching BERT performance with better training efficiency. NomicBERT (Nussbaum et al., 2024) and GTE-en-MLM (Zhang et al., 2024) (developed concurrently to this work) introduced longer-context encod...

- **Match 2** (sim 0.589) [body/sentence] Conclusion: ModernBERT has a native sequence length of 8,192 tokens and incorporates recent architecture improvements, such as GeGLU layers, RoPE positional embeddings, and alternating local-global attention.

- **Match 3** (sim 0.580) [body/paragraph] Downstream Results and Discussion: Short-Context Retrieval On BEIR, both variants of ModernBERT outperform existing encoders in both the DPR and ColBERT settings, including the recent GTE-en-MLM and NomicBERT models designed to serve as better backbones for retrieval (Zhang et al., 2024;Nussbaum et al., 2024) ModernBERT-large increases its lead despite having comparatively fewer parameters at 395M to GTE-en-MLM-large's 435M.

Reference [24]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2509.06888

Resolved title: mmBERT: A Modern Multilingual Encoder with Annealed Language Learning

Download status: already_indexed

Raw reference: [24] Marc Marone et al. mmBERT: A Modern Multilingual Encoder with Annealed Language Learning. 2025. arXiv: 2509.06888 [cs.CL]. url: <https://arxiv.org/abs/2509.06888>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.625 **hint:** body **zone:** body

Citing: Newer families such as XLM-V[22], ModernBERT[23], and mmBERT[24] aim to improve multilingual coverage, tokenizer behaviour, or runtime efficiency.

- **Match 1** (sim 0.625) [body/sentence] Conclusion: MMBERT improves over the previous generation of multilingual encoders such as XLM-R while being a drop-in replacement.
- **Match 2** (sim 0.621) [body/sentence] Introduction: MMBERT improves over XLM-R across the board, and even beats models like OpenAI's o3 (OpenAI, 2025) and Google's Gemini 2.5 Pro (Comanici et al., 2025) on low-resource languages.
- **Match 3** (sim 0.618) [body/paragraph] Introduction: MMBERT improves over XLM-R across the board, and even beats models like OpenAI's o3 (OpenAI, 2025) and Google's Gemini 2.5 Pro (Comanici et al., 2025) on low-resource languages. We show that including the low-resource languages in the decay phase enables rapid learning, boosting performance on these languages roughly 2x despite only using 100B tokens. Overall, MMBERT is the first model to show significant improvements over XLM-R for massively mu...

Reference [25]

Occurrences in work: 2

Resolved DOI: 10.1007/s10579-024-09759-3

Resolved title: PESTS: Persian_English cross lingual corpus for semantic textual similarity

Download status: already_indexed

Raw reference: [25] Mohammad Abdous, Poorya Piroozfar, and Behrouz MinaeiBidgoli. "PESTS: Persian_English cross lingual corpus for semantic textual similarity". In: Language Resources and Evaluation 59.2 (2025), pp. 1179–1199. issn: 15740218. doi: 10.1007/s10579-024-09759-3. url: <https://doi.org/10.1007/s10579-024-09759-3>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.632 **hint:** body **zone:** body

Citing: Recent studies report strong results for transformer-based multilingual embeddings in semantic comparison and paraphrase-oriented cross-lingual settings[25].

- **Match 1** (sim 0.632) [body/sentence] Conclusion: Semantic textual similarity represents a significant area of research in natural language processing, particularly in the context of cross-lingual applications.
- **Match 2** (sim 0.553) [body/paragraph] Related works: In their study, Tang et al. (Tang et al., 2018) developed a model for lowresource languages such as Spanish, Arabic, Indonesian, and Thai. Using a monolingual semantic similarity model framework, they extended it to multilingual mode. This approach demonstrated that through the utilization of a shared multilingual encoder, each sentence could exhibit distinct embeddings depending on the target language.
- **Match 3** (sim 0.546) [abstract/paragraph] abstract: In recent years, there has been significant research interest in the subtask of natural language processing called semantic textual similarity. Measuring semantic similarity between words or terms, sentences, paragraphs and documents plays an important role in natural language processing and computational linguistics. It finds applications in question-answering systems, semantic search, fraud detection, machine translation, information retrieval...

2. Label: related **sim:** 0.541 **hint:** unclear **zone:** body

Citing: This is suitable for cross-lingual verification because it does not rely on exact wording overlap and does not assume literal translation[25].

- **Match 1** (sim 0.541) [body/sentence] Related works: To predict an English word, this model can look at both English and French translations, and it seeks to align English and French embeddings.
- **Match 2** (sim 0.534) [body/sentence] Related works: In the domain of cross-lingual semantic textual similarity, the aim is to estimate the degree of the meaning similarity between two sentences, each in a different language (Hercig & Král, 2021).
- **Match 3** (sim 0.514) [body/paragraph] Related works: One potential strategy for addressing this challenge involves employing machine translation-based methods to translate sentences from low-resource languages like Persian into high-resource languages such as English. However, a major drawback of these approaches is the presence of errors in machine translation, and their effectiveness is strongly contingent upon the quality of the translation (Bjerva & Ostling, 2017). In our work, we leverage mul...

Reference [26]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [26] LLC Turnitin. Turnitin. 2025. url: <https://www.turnitin.com/> (visited on 09/21/2025).

Verification status: no_source_text

Reference [27]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [27] LLC Turnitin. iThenticate. 2025. url: <https://www.ithenticate.com/> (visited on 09/21/2025).

Verification status: no_source_text

Reference [28]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [28] Fakulta informatiky Masarykovy univerzity – Vývojový tým IS MU. Odevzdej CZ. 2025.
url: <https://odevzdej.cz/> (visited on 09/21/2025).

Verification status: no_source_text

Reference [29]

Occurrences in work: 2

Download status: no_valid_doi

Raw reference: [29] CVTI SR. Centrálny register záverečných a kvalifikačných prác. 2025. url: <https://crzp.cvtisr.sk/> (visited on 09/21/2025).

Verification status: no_source_text

Reference [30]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/n19-1361

Resolved title: Structural Scaffolds for Citation Intent Classification in Scientific Publications

Download status: downloaded

Downloaded PDF: pdfs\N19-1361.pdf

Raw reference: [30] Arman Cohan et al. "Structural Scaffolds for Citation Intent Classification in Scientific Publications". In: Proceedings of the 2019 Conference of the References North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). Ed. by Jill Burstein, Christy Doran, and Thamar Solorio. Minneapolis, Minnesota: Association for Computational Linguistics, June 2019, pp. 3586–3596. doi: 10.18653/v1/N19-1361. url: <https://aclant...>

Citing sentences and best evidence

1. Label: related **sim:** 0.573 **hint:** unclear **zone:** body

Citing: Some research focuses on citation intent and rhetorical function, as in SciCite[30].

- **Match 1** (sim 0.573) [body/section] Introduction: Citations play a unique role in scientific discourse and are crucial for understanding and analyzing scientific work (Luukkonen, 1992;Leydesdorff, 1998). They are also typically used as the main measure for assessing impact of scientific publications, venues, and researchers (Li and Ho, 2008). The nature of citations can be different. Some citations indicate direct use of a method while some others merely serve as acknowledging a prior work. The...

- **Match 2** (sim 0.573) [body/paragraph] Introduction: Citations play a unique role in scientific discourse and are crucial for understanding and analyzing scientific work (Luukkonen, 1992;Leydesdorff, 1998). They are also typically used as the main measure for assessing impact of scientific publications, venues, and researchers (Li and Ho, 2008). The nature of citations can be different. Some citations indicate direct use of a method while some others merely serve as acknowledging a prior work. The...

- **Match 3** (sim 0.544) [body/sentence] Structural scaffolds: In scientific writing there is a connection between the structure of scientific papers and the intent of citations.

Reference [31]

Occurrences in work: 2

Resolved DOI: 10.48550/arxiv.2511.16198

Resolved title: SemanticCite: Citation Verification with AI-Powered Full-Text Analysis and Evidence-Based Reasoning

Download status: already_indexed

Raw reference: [31] Sebastian Haan. SemanticCite: Citation Verification with AI-Powered FullText Analysis and Evidence-Based Reasoning. 2025. arXiv: 2511 . 16198 [cs.CL]. url: <https://arxiv.org/abs/2511.16198>.

Citing sentences and best evidence

1. Label: related **sim:** 0.578 **hint:** unclear **zone:** body

Citing: An even more direct systems-level point of comparison is Haan's[31] SemanticCite, which proposes citation verification through AI-powered full-text analysis and evidence-based reasoning.

- **Match 1** (sim 0.578) [body/paragraph] Implications for Research Quality and Broader Applications: The demonstrated effectiveness of SemanticCite has broader implications for research quality and scholarly communication. Automated citation verification tools can significantly enhance peer review efficiency by identifying problematic citations before human review, allowing reviewers to focus on higher-level scientific assessment rather than time-consuming fact-checking.

- **Match 2** (sim 0.563) [body/section] Contributions: This paper introduces SemanticCite, the first comprehensive AI-powered framework for automated fulltext citation verification. Unlike existing approaches that rely on abstract-level analysis or binary classification schemes, our system performs deep semantic analysis of complete source documents using a 4-class taxonomy that captures the nuanced relationships between citations and their references. SemanticCite bridges the gap between automated...

- **Match 3** (sim 0.563) [body/paragraph] Contributions: This paper introduces SemanticCite, the first comprehensive AI-powered framework for automated fulltext citation verification. Unlike existing approaches that rely on abstract-level analysis or binary classification schemes, our system performs deep semantic analysis of complete source documents using a 4-class taxonomy that captures the nuanced relationships between citations and their references. SemanticCite bridges the gap between automated...

2. Label: supported **sim:** 0.632 **hint:** body **zone:** body

Citing: The system must work with full-text documents, identify citations, retrieve and analyse source material, and produce evidence that can justify the verification result[31].

- **Match 1** (sim 0.632) [body/sentence] System Architecture: The system processes reference documents through text extraction and chunking, whilst citations undergo preprocessing to remove attribution markers and extract core factual claims.

- **Match 2** (sim 0.627) [body/sentence] Research Objectives: First, we develop an automated full-text citation verification system that transcends surface-level bibliographic checks to perform deep semantic analysis of citation content against complete source documents.

- **Match 3** (sim 0.623) [body/sentence] Retrieval-Augmented Generation: In citation verification contexts, RAG enables systems to retrieve relevant evidence from source documents and generate informed assessments of citation accuracy.

Reference [32]

Occurrences in work: 1

Resolved DOI: 10.1287/inte.1070.0317

Resolved title: The Ombudsman: Verification of Citations: Fawltly Towers of Knowledge?

Download status: already_indexed

Raw reference: [32] Malcolm Wright and J. Scott Armstrong. "The Ombudsman: Verification of Citations: Fawltly Towers of Knowledge?" In: Interfaces 38.2 (Apr. 2008), 125–139. issn: 1526-551X. doi: 10.1287/inte.1070.0317. url: <http://dx.doi.org/10.1287/inte.1070.0317>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.765 **hint:** body **zone:** body

Citing: Wright and Armstrong[32] described citation accuracy as a structural weakness in scholarly communication.

- **Match 1** (sim 0.765) [body/sentence] Brian Martin: Wright and Armstrong are concerned primarily with faulty citation as a factor in poor research.
- **Match 2** (sim 0.672) [body/paragraph] Conclusions: Wright and Armstrong are right! There is a substantial citation problem, which must be corrected. As Wright and Armstrong emphasize, the problem includes inaccurate use of procedures described in the publication being cited. In addition, errors in citations are also disconcerting, although in some cases they may be humorous, e.g., I was cited as authoring a book written 34 years before I was born! Their paper caused me to reflect on why journals...
- **Match 3** (sim 0.656) [body/sentence] Redundant Citations: An insidious problem, which Wright and Armstrong do not address, is that of spurious and redundant citations.

Reference [33]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2402.14268

Resolved title: Can Large Language Models Detect Misinformation in Scientific News Reporting?

Download status: already_indexed

Raw reference: [33] Yupeng Cao et al. Can Large Language Models Detect Misinformation in Scientific News Reporting? 2025. arXiv: 2402.14268 [cs.CL]. url: <https://arxiv.org/abs/2402.14268>.

Citing sentences and best evidence

1. Label: related **sim:** 0.612 **hint:** unclear **zone:** body

Citing: Large language models are now also being tested in neighbouring tasks such as scientific-misinformation detection[33], but these approaches still raise questions of reproducibility, cost, and consistency.

- **Match 1** (sim 0.612) [body/paragraph] INTRODUCTION: Several websites are maintained by science reporters (Health News Review 3) and scientists (Science Feedbackfoot_1) to track scientific misinformation in the media. Although such manual debunking is important, the sheer volume of scientific news can make this task unscalable. Natural language processing (NLP) based approaches have consequently started to emerge to deal with this problem. These methods typically involve language analysis, like ...

- **Match 2** (sim 0.599) [body/paragraph] INTRODUCTION: While these works have laid the foundation to address this problem, the area is still in its nascence. Several challenges remain unaddressed: 1) there are no existing taxonomies to define dimensions of the scientific validity of scientific news articles that can be used in automated methods to detect misinformation in scientific news; 2) all existing datasets for scientific fact-checking relies on explicit claim generation from the news articles...

- **Match 3** (sim 0.577) [body/section] RELATED WORK: As mentioned earlier, scientific misinformation detection is still in its nascence and while related to misinformation detection in general, it is a harder problem since the language characteristics of the scientific communication is different from the formal format of scientific publications. The problem of scientific misinformation is related to two other concepts in NLP, including: 1) fact-checking (claim verification) 2) scientific language ...

Reference [34]

Occurrences in work: 1

Resolved DOI: 10.1080/0361526x.2024.2433256

Resolved title: The Problem with False Positives: AI Detection Unfairly Accuses Scholars of AI Plagiarism

Download status: already_indexed

Raw reference: [34] Louie Giray. "The Problem with False Positives: AI Detection Unfairly Accuses Scholars of AI Plagiarism". In: The Serials Librarian 85.5-6 (2024), pp. 181–189. doi: 10.1080/0361526X.2024.2433256. eprint: <https://doi.org/10.1080/0361526X.2024.2433256>. url: <https://doi.org/10.1080/0361526X.2024.2433256>.

Citing sentences and best evidence

1. Label: related **sim:** 0.553 **hint:** unclear **zone:** body

Citing: Recent work also warns that threshold choices and dataset bias can distort reported performance, especially in imbalanced settings[34][35].

- **Match 1** (sim 0.553) [body/sentence] Algorithmic biases: These biases, often baked into the algorithms, can unfairly target specific writing styles or even cultural differences in language use.
- **Match 2** (sim 0.501) [body/section] Algorithmic biases: One of the biggest issues with AI detection tools is the presence of algorithmic biases. These biases, often baked into the algorithms, can unfairly target specific writing styles or even cultural differences in language use. For instance, a detection tool might be more likely to flag work written by non-native English speakers, simply because their sentence structures differ from standard English conventions. This bias mirrors what Steele and A...
- **Match 3** (sim 0.501) [body/paragraph] Algorithmic biases: One of the biggest issues with AI detection tools is the presence of algorithmic biases. These biases, often baked into the algorithms, can unfairly target specific writing styles or even cultural differences in language use. For instance, a detection tool might be more likely to flag work written by non-native English speakers, simply because their sentence structures differ from standard English conventions. This bias mirrors what Steele and A...

Reference [35]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2505.20158

Resolved title: Evaluating Software Plagiarism Detection in the Age of AI: Automated Obfuscation and Lessons for Academic Integrity

Download status: already_indexed

Raw reference: [35] Timur Sa■lam and Larissa Schmid. Evaluating Software Plagiarism Detection in the Age of AI: Automated Obfuscation and Lessons for Academic Integrity. 2025. arXiv: 2505.20158 [cs.SE]. url: <https://arxiv.org/abs/2505.20158>.

Citing sentences and best evidence

1. Label: related **sim:** 0.553 **hint:** unclear **zone:** body

Citing: Recent work also warns that threshold choices and dataset bias can distort reported performance, especially in imbalanced settings[34][35].

- **Match 1** (sim 0.553) [body/sentence] Internal Validity: Public datasets often contain incomplete or biased plagiarism labels.

- **Match 2** (sim 0.531) [body/sentence] Similarity Metrics: Due to varying similarity distributions for different datasets, they can only be set after seeing the results, thus introducing a strong bias.

- **Match 3** (sim 0.528) [body/paragraph] Similarity Metrics: A common anti-pattern in existing works is evaluating plagiarism detectors using a fixed similarity threshold where scores above count as successful detections, and those below do not. While this simplifies deriving precision, recall, and F1 scores, this approach is fundamentally flawed, as the threshold can arbitrarily influence results and it can be tuned to favor one approach. Since no universal threshold fits all datasets, thresholds are ch...

Reference [36]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [36] Crossref. Crossref. 2026. url: <https://www.crossref.org/>.

Verification status: no_source_text

Reference [37]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [37] GROBID. <https://github.com/kermitt2/grobid>. 2008–2026. swb: 1: dir:dab86b296e3c3216e2241968f0d63b68e8209d3c.

Verification status: no_source_text

Reference [38]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2006.07264

Resolved title: Low-resource Languages: A Review of Past Work and Future Challenges

Download status: already_indexed

Raw reference: [38] Alexandre Magueresse, Vincent Carles, and Evan Heetderks. Low-resource Languages: A Review of Past Work and Future Challenges. June 2020. doi: 10.48550/arXiv.2006.07264.

Citing sentences and best evidence

1. Label: related **sim:** 0.617 **hint:** unclear **zone:** abstract

Citing: Another weakness is its dependence on pretrained multilingual models, whose performance may be lower for low-resource languages or for highly specialised academic language[38].

- **Match 1** (sim 0.617) [abstract/sentence] abstract: A current problem in NLP is massaging and processing low-resource languages which lack useful training attributes such as supervised data, number of native speakers or experts, etc.

- **Match 2** (sim 0.593) [body/section] Introduction: In the 1990s, the tools of Natural Language Processing (NLP) experienced a major shift, transitioning from rule-based approaches to statisticalbased techniques. Most of today's NLP research focuses on 20 of the 7000 languages of the world, leaving the vast majority of languages understudied. These languages are often referred to as low-resource languages (LRLs), ill-defined as this qualifier can be. LRLs can be understood as less studied, resour...

- **Match 3** (sim 0.587) [body/sentence] Classification: From a machine learning standpoint, one alternative is adopting lighter, less resource-dependent baseline models that are still successful with the English language and modifying them to maximize success with a particular LRL.

Reference [39]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/d19-1371

Resolved title: SciBERT: A Pretrained Language Model for Scientific Text

Download status: already_indexed

Raw reference: [39] Iz Beltagy, Kyle Lo, and Arman Cohan. "SciBERT: A Pretrained Language Model for Scientific Text". In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing. 2019, pp. 3615–3620. doi: 10.18653/v1/D19-1371.

Citing sentences and best evidence

1. Label: related **sim:** 0.542 **hint:** unclear **zone:** body

Citing: Domain mismatch creates an additional problem because pretrained language models are often trained on general web data rather than academic prose[39].

- **Match 1** (sim 0.542) [body/sentence] Conclusion and Future Work: Because these language models are costly to train, we aim to build a single resource that's useful across multiple domains.

- **Match 2** (sim 0.484) [abstract/paragraph] abstract: Obtaining large-scale annotated data for NLP tasks in the scientific domain is challenging and expensive. We release SCIBERT, a pretrained language model based on BERT (Devlin et al., 2019) to address the lack of highquality, large-scale labeled scientific data. SCIBERT leverages unsupervised pretraining on a large multi-domain corpus of scientific publications to improve performance on downstream scientific NLP tasks. We evaluate on a suite of ...

- **Match 3** (sim 0.477) [body/paragraph] Introduction: As shown through ELMo (Peters et al., 2018), GPT (Radford et al., 2018) and BERT (Devlin et al., 2019), unsupervised pretraining of language models on large corpora significantly improves performance on many NLP tasks. These models return contextualized embeddings for each token which can be passed into minimal task-specific neural architectures. Leveraging the success of unsupervised pretraining has become especially important especially when t...

Reference [40]

Occurrences in work: 1

Resolved DOI: 10.47191/ijsshr/v7-i10-39

Resolved title: Challenges of Translating Idiomatic Expressions: A Cross-Linguistic Analysis at a University in Hanoi, Vietnam

Download status: already_indexed

Raw reference: [40] Dang Chung. "Challenges of Translating Idiomatic Expressions: A CrossLinguistic Analysis at a University in Hanoi, Vietnam". In: International Journal of Social Science and Human Research 07 (2024). doi: 10.47191/ ijsshr/v7-i10-39.

Citing sentences and best evidence

1. Label: supported **sim:** 0.628 **hint:** body **zone:** body

Citing: Similarity scores may be affected by translation differences, paraphrasing, domain-specific terminology, or idiomatic expressions[40].

- **Match 1** (sim 0.628) [body/sentence] I. INTRODUCTION 1.1.: As Wray (2018) points out, languages may not always have an exact idiomatic equivalent for a particular phrase, which forces translators to choose between paraphrasing, finding a comparable idiom, or explaining the meaning through non-idiomatic language.
- **Match 2** (sim 0.614) [body/sentence] Finding Equivalents in the Target Language: However, finding equivalents can be challenging when there are significant cultural or linguistic differences between the source and target languages.
- **Match 3** (sim 0.608) [body/sentence] VI. CONCLUSION: Linguistic factors, such as differences in syntax and idiomatic structure, coupled with cultural mismatches, contribute to the complexity of translating idioms accurately and meaningfully.

Reference [41]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.1908.10084

Resolved title: Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks

Download status: already_indexed

Raw reference: [41] Nils Reimers and Iryna Gurevych. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks". In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Nov. 2019. url: <http://arxiv.org/abs/1908.10084>.

Citing sentences and best evidence

1. Label: related **sim:** 0.477 **hint:** unclear **zone:** body

Citing: The GPU was used exclusively by the local embedding model paraphrase-xlm-r-multilingual-v1[41] via sentence-transformers.

- **Match 1** (sim 0.477) [body/paragraph] Model: SBERT adds a pooling operation to the output of BERT / RoBERTa to derive a fixed sized sentence embedding. We experiment with three pooling strategies: Using the output of the CLS-token, computing the mean of all output vectors (MEANstrategy), and computing a max-over-time of the output vectors (MAX-strategy). The default configuration is MEAN.

- **Match 2** (sim 0.453) [body/sentence] Related Work: -head attention over 12 (base-model) or 24 layers (large-model) is applied and the output is passed to a simple regression function to derive the final label.

- **Match 3** (sim 0.448) [body/sentence] Model: We experiment with three pooling strategies: Using the output of the CLS-token, computing the mean of all output vectors (MEANstrategy), and computing a max-over-time of the output vectors (MAX-strategy).